

Fire-Sale Spillovers in Debt Markets

ANTONIO FALATO, ALI HORTAÇSU, DAN LI, and CHAEHEE SHIN

ABSTRACT

Fire sales induced by investor redemptions have powerful spillover effects among funds that hold the same assets, hurting peer funds' performance and flows, and leading to further asset sales with negative bond price impact. A one-standard-deviation increase in our fire-sale spillover measure leads to a 45 (90) bp decrease in peer fund returns (flows) and a two percentage point increase in the likelihood of a large bond price drop. The results hold in a regression-discontinuity design addressing identification concerns. Timing, heterogeneity, instrumental-variable, and placebo tests further support the price-impact mechanism. Model-based counterfactual and stress-test analyses quantify the financial stability implications.

STARTING WITH THE INFLUENTIAL WORK of Shleifer and Vishny (1992), a major topic in finance is that basic equilibrium considerations may compound the cost of financial distress, thus aggravating the fragility of the financial system. Broadly, the idea is that distressed asset sales are particularly costly in circumstances when the pool of potential buyers of a specialized or illiquid asset is equally likely to be financially constrained and therefore unable to supply liquidity. More specifically, as Coval and Stafford (2007) note, “the cycle whereby capital flows can force widespread trading in individual securities, resulting in institutional price pressure, which in turn affects fund performance and eventually feeds back into capital flows, is intriguing” (p. 510).

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Though theoretically appealing, there has been no systematic attempt to test fire-sale spillovers empirically. In an attempt to fill this gap in the literature, we take the idea of fire-sale spillovers seriously and develop a novel approach to assess it empirically. We use rich microdata on flows, performance, and asset holdings for the universe of open-end fixed-income mutual funds. These data are an ideal laboratory to study fire-sale spillovers because bonds are relatively illiquid assets (Madhur et al. (2010), Bao, Pan, and Wang (2011)), which can give rise to liquidity mismatch or transformation for open-end funds that provide daily redemption rights to their investors (see, for example, Goldstein, Jiang, and Ng (2017)). We identify the peers of any given fund as those funds that hold the same assets. We present reliable evidence of powerful spillover effects among funds that hold the same assets, with fire sales induced by investor redemptions hurting peer funds' performance and flows, leading in turn to further asset sales that have a negative bond price impact.

Our study also contributes to the postcrisis debate regarding threats to financial stability coming from nonbank financial institutions.¹ Episodes of heightened credit market volatility such as the Taper Tantrum of the summer of 2013, the dramatic shift in credit intermediation away from banks and toward bond markets since the crisis, and the fact that funds have become one of the largest holders of corporate debt highlight the importance of better understanding their systemic consequences.²

The key measurement hurdles are to identify peer funds whose asset sales may affect others within the group and to address endogeneity concerns. Intuitively, a fund with high exposure to fire-sale spillovers holds a high proportion of assets that are fire-sold by other funds. To operationalize this measure, we build on the well-known "flow pressure" measure of Coval and Stafford (2007), which is constructed by looking at changes in holdings of a given security by mutual funds that experience large investor outflows or inflows. For each asset held by a given fund i , we calculate the Coval-Stafford pressure measure based on out/inflows experienced by funds *other than* i . The weighted sum (according to i 's portfolio weights) of this asset-specific flow pressure then yields our mea-

¹ See, for example, Feroli et al. (2014), the Financial Stability Oversight Council (FSOC) 2014 Notice Seeking Comment on Asset Management Products and Activities, 79 Fed. Reg. 77488 (December 24, 2014), the SEC request of public feedback on the Office of Financial Research (OFR) report (<http://www.sec.gov/comments/am-1/am-1.shtml>), and the 2014 Financial Stability Board (FSB) Consultation on "Assessment Methodologies for Identifying Non-Bank Non-Insurer Global Systemically Important Financial Institutions" (http://www.fsb.org/2014/01/r_140108-2/), all of which include consideration of stress testing for asset managers.

² Corporate debt trading volumes roughly doubled since the crisis from about \$14B in 2008 to over \$30B in 2017, and outstandings increased by roughly 50%. The asset management industry experienced fast growth in the fixed-income sector, with assets in open-end fixed-income funds growing from \$1.5 trillion in 2008 to \$3.5 trillion by the end of 2014, and net inflows exceeding \$1.3 trillion. Corporate bond funds also grew dramatically, with assets in open-end investment-grade (high-yield) mutual funds growing from about \$500 billion (\$100B) in 2008 to \$1.5 trillion (\$300B) in 2014. As a result, mutual funds have become major players in debt markets, with their market share—the ratio of their assets to total outstandings—growing from about 15% (10%) in 2008 to over 30% (20%) for investment-grade (high-yield) corporate bonds.

sure of fund i 's exposure to flow-induced asset sales/purchases by "peer" funds holding the same assets.

Our empirical approach allows us to trace the chain of events associated with price pressure from peer funds' flow-related trading. Specifically, we show that (i) outflow-induced asset sales have a negative price impact in the corporate bond market, (ii) there is a strong negative association between a fund's performance and the outflow-related asset sales of its peers, (iii) fund flows are also adversely affected by peer outflow-related asset sales, and (iv) there is a "second-round" price impact, with peer fire sales leading to further fire sales that have a negative bond price impact. The fire-sale spillover effect on other funds via exposure through common holdings is economically large.

For example, suppose that a given fund's monthly performance is flat, but its peers underperform, on average, by 3% in a given month. In this case, we estimate that the resulting one-standard-deviation (1-SD) increase in outflow-induced asset sales by peer funds will decrease the returns of a given fund by about 45 bps and increase its outflows by about 90 bps, which are roughly equal to the sample mean of returns and half the sample mean of fund flows, respectively. This second round of outflows will lead to an additional increase in the likelihood of a large decrease in bond prices of almost 2%.³ The relative magnitudes plausibly square together, with the estimates of the spillover effect corresponding to about one-quarter (one-tenth) of a standard deviation movement in fund returns (flows) and the estimate of the price impact corresponding to about one-third of a standard deviation movement in bond price changes. Moreover, the magnitudes of the spillover effects are large but plausible relative to those of the "first-round" effects, where peer funds experience outflows of about 4.5% and the resulting first round of fire sales increases the likelihood of a large bond price drop by about 8%.

An important concern with our measure is that it may erroneously pick up latent asset-specific shocks that are commonly observed by funds that choose to hold the same securities. To address this issue, which is an instance of the classic "reflection problem" discussed by Manski (1993), we use two main empirical strategies. First, inspired by Anton and Polk (2014), we construct our peer flow pressure measures based on funds' exposure to the 2003 mutual fund scandal. McCabe (2009) and Kisin (2011) show that funds of implicated fund families experienced significant outflows that were long-lasting, arguably due to reputation effects, but were not due to concerns about the quality of assets held by the funds. Second, we use a regression discontinuity (RD) strategy based on sharp changes in fund flows around the (un)assignment of Morningstar five-star ratings (as documented by, for example, Del Guercio and Tkac (2008) and Barrot (2016)). The source of identification here is that the sharp difference in (peer) fund flows observed for funds that fall slightly short of Morningstar's five-star performance cutoffs versus funds that are slightly above these cutoffs is likely not related to changes in common industry fundamentals. Importantly,

³ Large price drops are defined as observations in the bottom decile of their respective distributions.

the results all continue to hold when we focus on plausibly exogenous drivers of peer outflows/pressure.⁴ Taken together, we find strong evidence of fire-sale feedback effects in debt markets.

The economic mechanism that leads to spillovers is downward-sloping demand for corporate bonds, through which fire sales, that is, sales that are forced by redemptions, have a negative price impact. To the extent that purchasing a corporate bond requires the use of significant capital, any of the factors that sideline arbitrage capital can plausibly lead to downward-sloping demand for bonds. Several pieces of direct evidence support this fire-sale mechanism. First, the spillover effect is concentrated among bonds and funds that are illiquid and at times when the market is illiquid.⁵ The effect is robustly smaller and sometimes even insignificant for peer buy pressure and is lasting, though it reverses over the course of three to four quarters. These facts are all consistent with slow-moving capital making high-valuation bidders relatively scarce and leading to persistence of arbitrage opportunities (see, for example, Mitchell, Pedersen, and Pulvino (2007) and Duffie (2010)). Finally, in placebo tests, there is no spillover for asset sales that are not flow driven or for funds that operate in the same markets but do not have common holdings, both of which increase confidence in the price-impact mechanism relative to information and investor learning alternatives.

Fire-sale spillovers can affect financial market stability by amplifying the effect of an initial shock to fund flows that is otherwise unrelated to fundamental asset values and thus increasing volatility.⁶ To clarify the conditions and circumstances under which fire-sale spillovers increase volatility, we use our estimates to construct two measures of fund exposure to spillovers, systemicness and vulnerability, and we explore their relation with the volatility of fund and bond returns. Generally, these measures are higher when (i) the degree of overlap in bond holdings with other funds is higher and (ii) debt market conditions are such that forced sales have a larger price impact. We use regression analysis and a model-based calibration to illustrate these mechanisms. First, we use forecasting regressions of fund and bond return volatility similar to Greenwood and Thesmar (2011) to test whether exposure to fire-sale spillover forecasts volatility. Our measure of fund vulnerability to peer fire sales is positively correlated with one-quarter-ahead monthly volatilities of fund returns. For example, a 1-SD increase in fund vulnerability to fire-sale spillovers leads to an increase in annual volatility of about 0.6%, which is approximately one-third of the sample mean. A model-based calibration shows that spillovers

⁴ The results are also robust to several specification checks that include adding controls for bond and fund characteristics as well as lagged performance and flows.

⁵ The fact that we find larger effects for illiquid funds and during illiquid times offers additional reassurance that mechanical explanations based on omitted common factors are unlikely to be driving our results, as such explanations do not necessarily predict the same response across subsamples.

⁶ The mechanism that can potentially hinder financial stability is that, as we have shown, spillovers lead to additional outflows and further impact bond prices over and above the initial effect of a given adverse shock.

increased fund return volatility by up to one and a half times during the crisis, and continue to have a smaller but nonnegligible effect in recent years. Second, sector-wide and security-level averages of our measures of systemicness and vulnerability are positively correlated with the future volatility of an aggregate index of bond fund returns and of individual bonds, respectively. Further corroborating the idea that fire-sale spillovers matter in the aggregate, shocks to even just the 10 most systemic fund families lead to large sector-wide outflows in a stress-test exercise for several alternative adverse scenarios.

Our paper makes two main contributions. First, we contribute to the literature on fire sales in finance (see Shleifer and Vishny (2011) for an overview) and more specifically in the asset management industry, which has shown convincing evidence regarding the price impact of forced asset sales.⁷ Coval and Stafford (2007) find evidence of stock price impact based on abnormal stock returns in a 12-month window around large mutual fund sales. Khan et al. (2012) and Jotikatschira et al. (2012) use similar data in a U.S. and international setting, respectively, and confirm the finding of significant price impact. Ellul et al. (2011) and Feldhütter (2012) document evidence of significant price pressure around downgrades for bonds with high exposure to insurance companies' fire-sale risk and examine the price difference between small trades and large trades on the same bond, respectively. Our contribution to this literature is to provide, to the best of our knowledge, the first systematic study of the spillover effect of fire sales induced by *peer* fund flows. That spillovers may result from price impact is clear conceptually from theory, but whether the price impact is strong enough to matter for other funds remains an open empirical question. As such, our result that fire sales are strong enough to spill over onto other funds that hold the same securities is new to the literature.⁸ The result is complementary to recent evidence in Chernenko and Sunderam (2020) that exposure to fire sales affects the liquidity policies of equity funds, and it is important because it supports feedback mechanisms that are central in the narrative of the crisis and in the theory of fire-sales starting from Shleifer and Vishny (1992).

Second, there is a growing literature on vulnerability of financial institutions and stability in financial networks, which has been mostly theoretical and focused on banks (Acemoglu, Ozdaglar, and Tahbaz-Salehi (2012), Acemoglu, Akcigit, and Kerr (2015), Di Maggio and Tahbaz-Salehi (2015), Greenwood, Landier, and Thesmar (2015), Egan, Hortacsu, and Matvos (2017)).⁹ A handful of recent papers have started to recognize the importance of run-like

⁷ This literature, in turn, builds on the well-established evidence of downward-sloping demand curves for single securities. For instance, Shleifer (1986) reports evidence of a price effect of index inclusion and, more recently, Chang, Hong, and Liskovich (2015) provide causal evidence using an RD design that exploits Russell 2000 index inclusion.

⁸ Our direct evidence that flow-induced *trading* has a spillover effect is also complementary to earlier evidence of flow-based predictability of equity fund performance in Lou (2012).

⁹ There is also a recent related literature on measuring interconnectedness for banks and insurance companies (Billio et al. (2012), Ellul et al. (2018), Girardi et al. (2018), and Duarte and Eisenbach (2019)). Our findings contribute to this literature by broadening its focus to flow-driven

incentives in nonlevered nonbank financial intermediaries like fixed-income mutual funds. Goldstein, Jiang, and Ng (2017) emphasize that these funds exhibit a concave flow-performance sensitivity, with their outflows being more sensitive to bad performance than their inflows are to good performance, which can lead to instability (see also Chen, Goldstein, and Jiang (2010) and Feroli et al. (2014)). Our contribution to this literature is to introduce a mechanism, fire-sale spillovers, which had not yet been the subject of formal empirical testing, and to show that it is strong enough to matter not just for individual funds, but for the overall fixed-income fund sector, and not just for the sector, but more broadly for bond market valuations. As such, our work helps inform the current policy and academic debate by clarifying a specific source of vulnerability of nonbank financial intermediaries that is grounded in theory and by establishing its relevance in the context of debt markets, which is important given the spectacular growth of these markets since the crisis.¹⁰

I. Data, Measurement, and Research Design

In this section, we detail our sample construction procedure, our approach to measuring peer fund flow-related trading, and our research design to assess spillovers. Detailed variable definitions are provided in Section I of the [Internet Appendix](#).¹¹

A. Fund Flows, Returns, and Portfolio Holdings

The primary data for our analysis come from two standard sources. Monthly mutual fund flows and returns, as well as fund characteristics such

spillovers at mutual funds and corroborating the importance of measures of interconnectedness based on portfolio overlap.

¹⁰ Third, recent empirical research in finance and macroeconomics examines the link between investor sentiment and credit market volatility. Greenwood and Hanson (2013) show that periods of credit growth are associated with low future returns to credit investors, as well as bust periods when credit declines. Greenwood, Hanson, and Jin (2016) develop a model of the endogenous two-way feedback between credit market sentiment and credit market outcomes. López-Salido, Stein, and Zakrajšek (2017) show that fluctuations in credit market sentiment are closely tied to future movements in aggregate economic activity, which is consistent with the idea that changes in credit market valuations may drive business cycle volatility. While these papers provide convincing theory and evidence that credit market sentiment matters, the ultimate sources or determinants of sentiment are still relatively understudied. Our contribution is to highlight the role of correlated flows among interconnected institutions as a potentially important driver of credit market sentiment. As such, our analysis also contributes to the literature on the valuation consequences of ownership linkages among mutual funds (see, for example, Greenwood and Thesmar (2011) and Anton and Polk (2014)). Using quarterly data on equity mutual funds' holdings, Greenwood and Thesmar (2011) show that equity return volatility should be, and indeed is, higher for stocks that rank high based on their proposed measure of "fragility," which is high when ownership concentration is high or when mutual fund flows volatility and cross-correlation are high. Anton and Polk (2014) document that cross-sectional variation in (quarterly) common ownership by equity funds predicts higher four-factor abnormal return correlations, controlling for standard risk factors as well as the degree of common analyst coverage.

¹¹ The [Internet Appendix](#) may be found in the online version of this article.

as size (net assets), are from the Center for Research in Security Prices (CRSP) Survivorship-Bias-Free Mutual Fund Database. From the CRSP database, we retrieve information on the universe of open-end fixed-income U.S. funds, which leads to a sample of 330,429 fund share class-month observations for 3,666 (254) unique funds (families) between January 1998 and December 2014. Data on investment objectives and quarterly security-level fund holdings are from the Thomson Reuters/Lipper eMAXX fixed-income database, whose coverage starts in the first quarter of 1998.¹² When necessary for the analysis, we retrieve additional information on security-level corporate bond trading volume and liquidity from TRACE, prices from the Merrill Lynch database, and bond characteristics at issuance from Mergent Fixed Income Securities Database (FISD).

Our main dependent variables of interest are fund flows and performance. Mutual fund flows are estimated following the prior literature (e.g., Chevalier and Ellison (1997)), which is to define net flows of funds to mutual fund (share class) i in month t as the percentage growth of new assets,

$$Flow_{i,t} = (TNA_{i,t} - (1 + r_{i,t})TNA_{i,t-1})/TNA_{i,t-1},$$

where $TNA_{i,t}$ is the total net assets under management of fund i in month t and $r_{i,t}$ is the fund's return (net of fees and expenses) over the period.¹³ Mutual fund performance is measured using monthly fund returns.

B. Measuring Network Linkages among Funds

The key measurement challenge is to define network linkages across financial institutions. Our notion of a fire-sale spillover requires that several different owners of a given security experience capital withdrawals, which leads them to sell the security at depressed prices in an effort to cover redemptions. It is through this “downstream” price pressure channel that redemptions “upstream” at fund A may spill over onto those of fund B. Based on this notion, our spillover measure needs to capture two main features: (i) common asset ownership and (ii) intensity of flow-driven capital withdrawals at other funds. To capture these features, we use a new approach that is similar to the recent literature on networks in macroeconomics (e.g., Acemoglu, Akcigit and Kerr (2015)) and knowledge spillovers in the economics of innovation (e.g., Jaffe (1986), Bloom, Schankerman, and Van Reenen (2013)).

The basic idea behind our approach is that flow-induced price pressure is transferred between funds that are “exposed” to each other. For any given fund, the probability that flow pressure is transferred to another depends on the extent to which the two funds invest in the same assets, with the polar case of two

¹² Since we use holdings to construct our main peer variables, January 1998 is our earliest available date. Because we are interested in actively managed funds, we exclude from the analysis funds that CRSP identifies as index and exchange-traded funds.

¹³ As is also standard practice in the literature, fund flows are winsorized at the 1% and 99% percentiles to mitigate the influence of outliers.

focused funds that are invested exclusively in the same asset representing the highest exposure and that of two focused funds that are invested exclusively in two different assets representing the lowest. Spanning the continuum of intermediate cases when funds are invested in multiple assets that are sometimes overlapping, the expected fire-sale spillover from one fund to another is an aggregate of these transfers from multiple funds across multiple assets.

More specifically, we identify the peers of any given fund as those funds that hold the same assets. We construct the measure of fire-sale spillover, *PeerFlow Pressure*_{*i,t*}, for each fund *i* as the weighted sum of fire-sale pressure from other funds, with weights capturing “exposure.” To implement this definition, for each of the $N = 3,666$ funds in the sample and each month *t* we start with a vector $w_{i,t} = (w_{i1t}, \dots, w_{ibt}, \dots, w_{int})$ of its *n* portfolio share holdings, that is, the portfolio percentage share holdings of each fund *i* in each asset *b* in each period *t*. Using these portfolio share holdings as weights, for each fund we take a weighted sum of price pressure from other funds’ flow-related trading in each of the securities it holds,

$$Peer\ Flow\ Pressure_{i,t} = \sum_{b=1}^n Flow\ Pressure_{b,t}^{j \neq i} * w_{i,b,t-1}, \quad (1)$$

where, for each asset *b*, we define *Flow Pressure*_{*b,t*} similarly to Coval and Stafford (2007) as the proportion of flow-induced net buys by peer funds,

$$Flow\ Pressure_{b,t}^{j \neq i} = \frac{Flow\ Induced\ Buys_{b,t}^{j \neq i} - Flow\ Induced\ Sales_{b,t}^{j \neq i}}{Offering\ Value_b},$$

where $Flow\ Induced\ Buys_{b,t}^{j \neq i} = \sum_{j \neq i} (\max(0, \Delta Holdings_{j,b,t}) | Flows_{j,t} > Percentile(90th))$ and $Flow\ Induced\ Sales_{b,t}^{j \neq i} = \sum_{j \neq i} (\max(0, -\Delta Holdings_{j,b,t}) | Flows_{j,t} < Percentile(10th))$. To avoid mechanical correlation, we exclude own funds, $j = i$, from the calculation of flow pressure, so that for each fund we isolate fire-sale pressure from *other* funds.¹⁴ Intuitively, a fund with high exposure to fire-sale spillovers holds a high proportion of assets that are fire-sold by other funds.

To examine potential asymmetric effects of buy versus sell pressure, we also present results for two alternative versions of this measure, which are constructed as weighted sums of *Flow Induced Buys*_{*b,t*}^{*j*≠*i*} and *Flow Induced Sales*_{*b,t*}^{*j*≠*i*},

$$Peer\ Buy\ Pressure_{i,t} = \sum_{b=1}^n \frac{Flow\ Induced\ Buys_{b,t}^{j \neq i}}{Offering\ Value_b} * w_{i,b,t-1},$$

$$Peer\ Sell\ Pressure_{i,t} = \sum_{b=1}^n \frac{Flow\ Induced\ Sales_{b,t}^{j \neq i}}{Offering\ Value_b} * w_{i,b,t-1}.$$

¹⁴ Bond offering value is from Mergent FISD.

Finally, for consistency and comparability of the spillover estimates with those of the price impact, we only include corporate bonds in our peer pressure measures. Table 1 provides descriptive statistics of the time distribution of our sample (number of funds and fund families in Panel A) and our main explanatory and dependent variables (means, standard deviations, and p95–p5 ranges in Panel B). In line with previous studies, there is substantial heterogeneity in fund flows and performance, as well as in fund characteristics such as size. In addition, our *Peer Flow Pressure* measure displays significant variation across fund-years, with a standard deviation of 0.114. To ease exposition and help gauge economic significance, throughout the analysis we report results in standard deviation units, that is, for our main variables standardized by dividing each raw variable by its respective standard deviation.

C. Research Design and Identification Strategy

We assess fire-sale spillovers in the most basic terms possible: Are the flows and performance of a given fund sensitive to the fire-sale pressure that is induced by the flows of its peers? To that end, we examine the following main relation:

$$Y_{i,t} = \alpha + \beta \times \text{Peer Flow Pressure}_{i,t} + \gamma \times X_{i,t-1} + \eta_i + \lambda_t + \nu_{i,t}, \quad (2)$$

where the outcome variables $Y_{i,t}$ for fund i in month t include the fund's performance and net flows, the main variable of interest is the spillover measure $\text{Peer Flow Pressure}_{i,t}$ defined in equation (1), and $X_{i,t-1}$ is a vector of controls for standard fund characteristics, which include fund family size, lagged own-fund flows or performance, and expense ratio. In robustness analysis, we present results for a specification that includes the full vector of controls. In our main analysis, we address two important issues, unobserved heterogeneity and endogeneity. We take on a third potential issue, dynamics, in graphical analysis and in additional robustness tests that estimate a more dynamic specification with distributed or various lags of the main explanatory variable.

To address unobserved heterogeneity, in all specifications we control for fund fixed effects by including a full set of fund-specific dummies, η_i . The inclusion of fund effects ensures that the parameter of interest, β , which represents the impact of peer fund flow pressure, is estimated from within-fund time-series variation. We address endogeneity due to transitory common shocks, a potential confound that may be erroneously picked up by the spillover measure, in two ways. First, in all tests we report results for a specification that controls directly for sector-wide shocks that are common across funds by adding controls for time fixed effects with the inclusion of a full set of year-specific dummies, λ_t .¹⁵ The idiosyncratic error term, $\nu_{i,t}$, is assumed to be correlated within the fund and potentially heteroskedastic (Petersen (2006)). Second, we recognize

¹⁵ We experimented with including alternative larger sets of higher frequency dummies, either quarter specific or month specific, and found that both lead to only minor changes in our estimates.

Table I
Summary Statistics

This table presents summary statistics (means and medians) for our sample, which comprises the universe of fixed-income funds. The data span the period January 1998 to December 2014 and comprises 330,429 fund (share class level)-month observations (column (1), Panel A) for 3,666 unique funds and 254 unique fund families (columns (2) and (3), Panel A, respectively). Variable definitions are provided in Section I of the [Internet Appendix](#).

Panel A: Sample Distribution			
	Obs	Funds (Share Class Level)	Families
	(1)	(2)	(3)
1998	3,854	671	123
1999	8,289	759	131
2000	9,232	862	137
2001	11,121	1,022	147
2002	12,681	1,176	152
2003	13,482	1,269	153
2004	14,477	1,375	159
2005	15,173	1,510	167
2006	16,952	1,640	171
2007	19,370	1,842	177
2008	21,909	2,126	189
2009	25,073	2,354	193
2010	27,663	2,550	200
2011	30,791	2,860	218
2012	34,383	3,127	225
2013	35,634	3,341	241
2014	30,345	3,280	232
Tot.	330,429	3,666	254

Panel B: Summary Statistics			
	Mean	Std Dev	p95–p5
<i>Main Explanatory Variable(s):</i>			
Peer Flow Pressure $Sd_{i,t}$	0.014	1	2.62
Peer Buy Pressure $Sd_{i,t}$	0.046	1	2.44
Peer Sell Pressure $Sd_{i,t}$	0.036	1	2.28
Peer Perf. $Sd_{i,t}$	0.201	1	2.91
Peer Extreme Outperf $Sd_{i,t}$	0.114	1	2.48
Peer Extreme Underperf $Sd_{i,t}$	0.163	1	2.39
<i>Main Outcome Variables:</i>			
Monthly Return	0.005	0.019	0.063
Extreme Underperformance	0.097	0.296	1
Monthly Flows	0.021	0.086	0.226
Extreme Outflows	0.103	0.305	1
<i>Fund Controls:</i>			
Fund Size (log\$Mil)	4.122	2.295	7.605
Family Size (log\$Mil)	8.015	2.002	6.763
Expense ratio (%)	0.011	0.005	0.015
Retail Fund	0.612	0.487	1
Equity Fund (primarily)	0.173	0.378	1

that the endogeneity issue in our peer-effect context is an instance of the classical “reflection problem,” which refers to the concern that correlation with peer flow pressure may arise due to endogenous selection of funds into peer groups and may pick up latent shocks that are common to funds that hold the same securities. As such, including time effects helps ameliorate but does not fully resolve the challenge of endogenous selection.

As discussed by Manski (1993), the ideal natural experiment would randomly assign similar types of funds to different types of peer treatment status. In other words, for our estimates of β in the reduced-form model of equation (2) to be identified, we need to show that fund flows are significantly correlated with plausibly exogenous characteristics of their peers that affect peer flows. The identification challenge is to find events or factors that are relevant for peer fund flows but that are otherwise plausibly random with respect to fund i ’s flows and performance.

We design three “quasi-natural” experiments that are geared toward generating this random assignment. Specifically, we propose three peer treatment variables, which are constructed for each fund similar to *Peer Flow Pressure* as an exposure-weighted sum of either fund-specific shocks or (functions of) thresholds that affect peer flows but are otherwise plausibly independent from own flows,

$$Peer\ Treatment\ Pressure_{i,t} = \sum_{b=1}^n Treatment\ Pressure_{b,t}^{j \neq i} * w_{i,b,t-1}, \quad (3)$$

where for each asset we define $Treatment\ Pressure_{b,t}^{j \neq i}$ similarly to flow pressure as the proportion of treatment-induced net buys:

$$Treatment\ Pressure_{b,t}^{j \neq i} = \frac{Treatment\ Induced\ Buys_{b,t}^{j \neq i} - Treatment\ Induced\ Sales_{b,t}^{j \neq i}}{Offering\ Value_b},$$

where $Treatment\ Induced\ Buys_{b,t}^{j \neq i} = \sum_{j \neq i} (\max(0, \Delta Holdings_{j,b,t}) | Treatment_{j,t}=0)$ and $Treatment\ Induced\ Sales_{b,t}^{j \neq i} = \sum_{j \neq i} (\max(0, -\Delta Holdings_{j,b,t}) | Treatment_{j,t}=1)$. To avoid mechanical correlation, we again exclude own-fund treatment from the calculation of treatment pressure. Our identification strategy is to estimate a version of the reduced-form model of equation (2), where we replace the potentially endogenous spillover measure, *Peer Flow Pressure* $_{i,t}$, with its plausibly exogenous counterpart, *Peer Treatment Pressure* $_{i,t}$. We consider three treatment variables, which are detailed in Section IV.

II. Motivating Evidence and Baseline Results

In this section, we present our baseline results on the chain of events that is associated with fire-sales spillovers and provide direct evidence in support of the price-impact mechanism. The key steps start with the price impact of bond fire sales, followed by the spillover effect of peer fire sales on fund performance, flows, and bond valuations, which shows that fire-sale externalities matter not

just for funds but more broadly for debt markets. To motivate the analysis, we provide evidence of predictability of own-fund performance by peer funds' performance, in that the performance of peer funds has strong predictive power up to one quarter ahead for the performance of any given fund that holds the same bonds. Predictability is a key reduced-form implication of fire-sale spillovers, because a fund's performance is likely to be harmed if the underperformance of its peers leads to capital withdrawals that depress its asset valuations via fire sales.

A. Motivating Evidence and Bond Price Impact

Table II summarizes our motivating evidence. We present estimates from predictability regressions of up to one-quarter-ahead fund performance (Panel A) and extreme underperformance (Panel B) as the outcome variables, and *Peer Fund Performance* as the explanatory variable of interest. Analogous to *Peer Flow Pressure*, peer performance is defined as a weighted sum of peer funds' returns using portfolio holdings as "exposure" weights and the (average) performance of the holders of a given security instead of flow-related trading to sum over securities. Even after controlling for past own-fund performance, there is a strong positive and highly statistically significant association between a fund's future performance and the past performance of its peers (Panel A, columns (1) to (3)). This evidence of predictability is a first indication that fire-sale spillovers are likely to be important, as poor performance of funds with overlapping holdings tends to be followed by poor performance of a given fund. Further supporting fire-sale spillovers, predictability is even stronger during illiquid times (columns (4)–(6)) and continues to hold for large underperformance (Panel B).

Before establishing fire-sale spillovers, we start by assessing whether asset sales under fund flow pressure bear the hallmark of costly fire sales, that is, involve securities whose valuations are depressed. Price impact of flow-related trading is the first step of the causal chain that leads to spillovers. The results of our bond event study are summarized in Table III, which reports the main results of regressions of monthly (changes in log) bond prices on *Flow Pressure*_{*b,t*} for a baseline specification that controls for bond (CUSIP) effects (Panel A). The coefficient on *Flow Pressure* in column (1) is positive and highly statistically significant, indicating that sales by mutual funds that are experiencing outflow pressure tend to harm bond valuations. The bond valuation effect is economically significant, as the coefficient estimate of about one percentage point (pp) in column (1) corresponds to a full quartile movement in the distribution of bond price changes.

In the next two columns we examine whether there is heterogeneity in the valuation effect. We expect that the valuation effect should be more negative whenever liquidation costs are higher. To explore this possibility, we repeat our analysis in subsample splits based on a cross-sectional (bond-level) proxy and a time-series (aggregate) proxy for illiquidity and fire-sale costs. The cross-sectional proxy for illiquidity in column (2) is lagged bond spreads

Table II
Motivating Evidence

This table reports predictability regressions of monthly fund returns (Panel A) and extreme underperformance (a dummy for returns in the bottom decile of the distribution, Panel B) up to a quarter ahead on peer fund performance. The independent variable in Panel A, *Peer Performance*, is a weighted sum of peer funds' returns, with weights calculated based on the asset allocation of a given fund. Specifically, we start with each security with information on holdings available from eMAXX and combine it with information on fund performance from CRSP to construct this measure for each fund by taking the weighted sum over the fund's securities holdings of the average return of other funds that hold a given security, with weights equal to the fund's percentage portfolio share holding of each security. The independent variable in Panel B is *Peer Extreme Underperformance*, which is a weighted sum of dummies for *Peer Performance* in the highest 10% of the distribution. To ease interpretation, these variables are expressed in standard deviation units. We report the baselines estimates in columns (1) to (3) and results for the crisis subsample in columns (4) to (6). The time period is 1998 to 2014. All specifications include fund fixed effects. Standard errors are clustered by fund. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively. Variable definitions are provided in Section I of the [Internet Appendix](#).

Panel A: Descriptive Analysis of Predictability of Fund Performance, $Y = \text{Monthly Return}_{i,t+N}$ (pct)						
	All Funds			Illiquid Times		
	$N = +1$ (1)	$N = +2$ (2)	$N = +3$ (3)	$N = +1$ (4)	$N = +2$ (5)	$N = +3$ (6)
Peer Perf. $Sd_{i,t}$	0.57*** (0.01)	0.78*** (0.01)	0.65*** (0.01)	0.58*** (0.02)	1.56*** (0.02)	1.03*** (0.03)
Fund Perf Sd_t	0.28*** (0.00)	-0.13*** (0.01)	-0.10*** (0.01)	0.68*** (0.01)	-0.31*** (0.01)	0.19*** (0.02)
FE	Fund	Fund	Fund	Fund	Fund	Fund
N	327,418	325,005	323,120	40,561	40,059	39,612
R ² (%)	11.04	14.26	15.27	22.40	24.38	25.17

(Continued)

Table II—Continued

Panel B: Descriptive Analysis of Predictability of Fund Performance, Y = Extreme Underperformance _{<i>t</i>+N} (pct)						
	All Funds			Illiquid Times		
	N = +1 (1)	N = +2 (2)	N = +3 (3)	N = +1 (4)	N = +2 (5)	N = +3 (6)
Peer Extreme Underperf. Sd _{<i>t</i>,<i>t</i>}	8.97*** (0.06)	9.72*** (0.07)	8.63*** (0.08)	5.17*** (0.19)	9.04*** (0.22)	8.31*** (0.26)
Fund Extreme Underperf Sd _{<i>t</i>}	4.59*** (0.20)	-4.42*** (0.28)	-0.86** (0.35)	11.28*** (0.52)	-6.72*** (0.75)	-18.69*** (0.97)
FE	Fund	Fund	Fund	Fund	Fund	Fund
N	327,418	325,005	323,120	40,561	40,059	39,612
R ² (%)	14.26	16.31	17.29	22.89	24.76	25.51

Table III
Analysis of Bond Price Impact

This table reports regressions of monthly (changes in log) bond prices on a variable that proxies for trades under fund flow (net) pressure. Panel A shows results for a flow-pressure variable constructed as flow-related net buy pressure using fund flows as in Coval and Stafford (2007). Specifically, we start with each security with information on holdings available from eMAXX and combine it with information on fund flows from CRSP to construct *Flow Pressure* as the sum of the value of net “forced” fund buys (buys by funds with large inflows buys minus sells by funds with large outflows relative to the offering value of the bond), with “forced” buys and sales defined based on the top and bottom deciles of the distribution of fund flows, respectively. Panel B show results for buy and sell pressure separately to allow for asymmetries. To ease interpretation, these variables are expressed in standard deviation units. We report results for the overall sample (columns (1) and (4)), for subsamples of illiquid bonds and in the crisis (columns (2) to (3) and (5) to (6), Panel A), for a specification with time (year) effects, and for a matched-sample estimator based on bond offering value, maturity, and lagged spread. Columns (4) to (6) limit the sample to the top versus bottom deciles of the flow-pressure variable to exclude either no trades or no trades by institutions under flow pressure as recommended by Khan, Kogan, and Serafeim (2012). The time period is 1998 to 2014. All specifications include controls for bond or time effects. Standard errors are clustered by bond. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively. Variable definitions are provided in Section I of the [Internet Appendix](#).

Panel A: Replication of Coval and Stafford (2007) – Monthly % Bond Price Changes (pct)						
	Baseline			Top and Bottom Deciles Only		
	All Bonds (1)	Illiquid Bonds (2)	Illiquid Times (3)	All Bonds (4)	Illiquid Bonds (5)	Illiquid Times (6)
Flow Pressure $Sd_{b,t}$	1.04*** (0.11)	3.25*** (0.37)	9.00*** (0.90)	1.01*** (0.17)	3.04*** (0.40)	7.81*** (1.37)
FE	Bond	Bond	Bond	Bond	Bond	Bond
N obs	429,449	83,156	47,724	76,809	30,651	9,573
N Bonds	10,880	4,698	3,008	7,295	3,246	1,693
R ² (%)	5.20	8.15	8.70	13.15	17.51	20.44
Sd LHS	4.99	9.33	9.38	7.12	9.65	12.63
IQR LHS	1.93	3.68	3.33	2.33	3.44	4.31
Panel B: Asymmetry & Robustness– Monthly % Bond Price Changes (pct)						
	Baseline			Top and Bottom Deciles Only		
	Asymmetry (1)	Time FE (2)	Matched Sample (3)	Asymmetry (4)	Time FE (5)	Matched Sample (6)
Buy Pressure $Sd_{b,t}$	0.75*** (0.19)	0.13 (0.15)	0.12 (0.18)	0.40 (0.33)	0.34 (0.22)	0.13 (0.31)
Sell Pressure $Sd_{b,t}$	−1.71*** (0.19)	−1.05*** (0.16)	−0.78*** (0.17)	−1.78*** (0.32)	−0.77*** (0.23)	−0.98*** (0.30)
FE	Bond	Time	Bond	Bond	Time	Bond
N obs	429,449	429,449	429,449	76,809	76,809	76,809
R ² (%)	5.21	3.88	5.61	13.15	4.18	13.37

over comparable-maturity Treasuries. We repeat our analysis in the subsample of relatively illiquid bonds, which are defined as those in the top quartile of the distribution of lagged spreads. The macro proxy for illiquidity in column (3) is defined as a dummy for the financial crisis.¹⁶ The results support a costly fire-sale interpretation, as the estimated coefficients are much larger in both columns, indicating that sales by mutual funds that experience outflow pressure tend to harm valuations more for more illiquid bonds and at times when debt markets are relatively illiquid. In columns (4) to (6) we verify that the baseline and heterogeneity results continue to hold if we exclude bond-quarters in which there are either no trades or no trades by institutions under flow pressure as recommended by Khan, Kogan, and Serafeim (2012).

Panel B provides further support for the fire-sale mechanism. Specifically, we repeat the analysis with *Flow Induced Buys*_{*b,t*} and *Flow Induced Sales*_{*b,t*} to allow for potentially asymmetric effects of buy versus sell pressure. The coefficient estimates in column (1) are statistically and economically significant for both variables, but the valuation effect is clearly asymmetric, with the negative effect of sale pressure being much larger than the positive effect of buy pressure. There is even stronger evidence of asymmetry in the alternative specification that controls for time effects by including month dummies (column (2)) and in the matched-sample analysis that considers the valuation effect relative to similar bonds based on bond offering value, bond maturity, and previous spread (column (3)), as in both alternatives only the coefficient estimate on sale pressure remains significant.

Finally, graphical analysis in Figure 1, Panel A, and additional regression analysis in Table IA.I, Panels A and B, examine the timing of the price impact in an attempt to better distinguish flow-related trading from information- or fundamentals-driven interpretations. In line with flow-related trading, the bulk of the valuation effect is concentrated in the immediate quarter when the sales happen and the following quarter, but the effect eventually reverses over the course of the third and fourth quarters. Taken together, the evidence from our bond event study shows that outflows-induced sales of corporate bonds have large and lasting, though not permanent, valuation consequences.¹⁷ In addition to being an internal validity check for our mechanism, these results replicate the approach and evidence of Coval and Stafford (2007) in the corporate bond market.

B. Fund Performance and Flows

The next three tables present our evidence on fire-sale spillovers. In Table IV, we summarize results from estimating equation (2) with fund

¹⁶ Brunnermeier and Pedersen (2009) show that asset market liquidity comoves with the funding liquidity of financial institutions that supply liquidity to asset markets.

¹⁷ Figure IA.1 shows a nonparametric version of the price path, which is in line with the results discussed in the main text. Table IA.I, Panel B, shows that none of the coefficient estimates on various leads of the flow pressure variable is significant, indicating the absence of statistically significant pre-event trends.

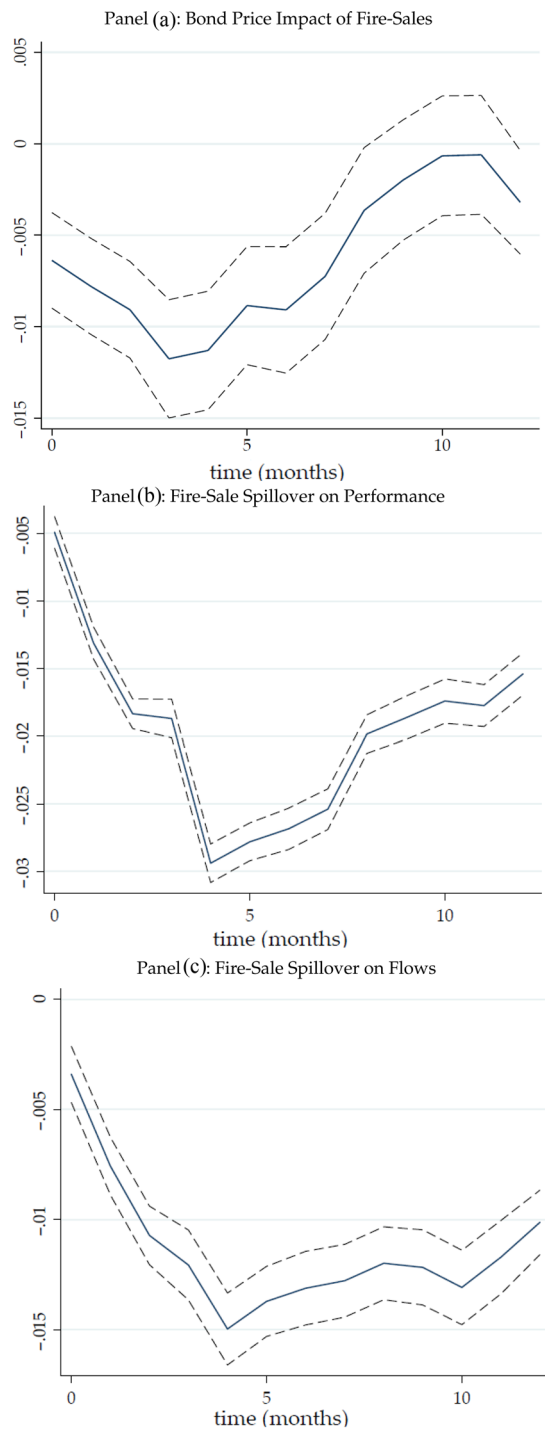


Figure 1. Timing of bond price impact and fire-sale spillovers. This figure reports results of calendar-time analysis of fire-sale spillovers. Panel (a) reports results of calendar-time

performance (returns, columns (1)–(3)) and extreme underperformance (a dummy for returns in the bottom decile of the distribution, columns (4)–(6)) as the outcome variables. The explanatory variables of interest are *Peer Flow Pressure* in Panel A and *Peer Buy Pressure* and *Peer Sell Pressure* in Panel B (see equation (1) for definitions). The coefficient on *Peer Flow Pressure* is positive (negative) and highly statistically significant for fund returns (extreme underperformance; Panel A, columns (1) and (4)), in line with the key mechanism behind fire-sale spillovers—capital withdrawals from peer funds harm own-fund performance by depressing asset valuations. Finally, the spillover effect is larger for relatively illiquid funds (those whose Lipper asset class is high yield, columns (2) and (5)) and during relatively illiquid times (during the financial crisis, columns (3) and (6)), and remains strongly significant when we address omitted variables and dynamics by adding controls for time effects (Panel B, columns (2) and (5)) and for standard fund-level covariates such as fund and family size, expense ratios, and lagged performance (Panel B, columns (3) and (6)).

Table V presents the estimates from equation (2) using fund flows (columns (1)–(3)) and extreme outflows (a dummy for fund flows in the bottom decile of the distribution, columns (4)–(6)) as the outcome variables. Again, the explanatory variables of interest are *Peer Flow Pressure* in Panel A and *Peer Buy Pressure* and *Peer Sell Pressure* in Panel B. We find a strong positive and highly statistically significant association between a fund's flows and *Peer Flow Pressure* (Panel A, column (1)), which is broadly supportive of the presence of strategic complementarity between own- and peer-fund flows. The result is stronger when illiquidity is higher (Panel A, columns (2), (3), (5), and (6)) and for peer sell pressure relative to peer buy pressure (Panel B, columns (1) and (4)) and is robust to controlling for common shocks by including time effects (Panel B, columns (2) and (5)) and for omitted variables and dynamics

analysis of the bond price impact in the overall sample by showing cumulative abnormal returns of a buy-and-hold strategy based on the top and bottom deciles of a flow pressure variable constructed as flow-related net buy pressure using fund flows as in Coval and Stafford (2007). Specifically, the panel plots coefficient estimates of a distributed lags version (with 12 lags) of the baseline specification for the bond price impact in Table III (column (4), Panel A) on $Net - SalePressure_{b,t} = \frac{Flow\ Induced\ Sales_{b,t} - Flow\ Induced\ Buys_{b,t}}{Offering\ Value_b}$ to ease interpretation. Panel B examines the cumulative effect of spillovers on fund performance by plotting coefficient estimates of a distributed lags version (with 12 lags) of the baseline specification for fund performance in Table IV (column (4), Panel A), where the explanatory variable of interest is *Peer Flow Pressure*, and the dependent variable, *Extreme Underperformance*, is defined as a dummy that takes value of one for fund-month observations in the bottom decile of the distribution of fund returns. Panel C examines the cumulative effect of spillovers on fund flows by plotting coefficient estimates of a distributed lags version (with 12 lags) of the baseline specification for fund flows in Table V (column (4), Panel A), where the explanatory variable of interest is *Peer Flow Pressure* and the dependent variable, *Extreme Outflows*, is defined as a dummy that takes value of one for fund-month observations in the bottom decile of the distribution of fund flows. Dotted lines denote 95% confidence intervals. See Section I of the Internet Appendix for detailed variable definitions and the legends of the respective tables for other specification details. (Color figure can be viewed at wileyonlinelibrary.com)

Table IV
Analysis of Fund Performance

This table reports regressions of monthly fund returns (columns (1)–(3)) and extreme underperformance (a dummy for returns in the bottom decile of the distribution, columns (4)–(6)) on peer fund flows (net) pressure. In Panel A, the independent variable, *Peer Flow Pressure*, is a weighted sum of peers' flow pressure, with weights calculated based on the asset allocation of a given fund. Specifically, we start with each security with information on holdings available from eMAXX and combine it with information on fund flows from CRSP to construct this measure for each fund-period by taking the weighted sum over the fund's securities holdings of the security-by-security pressure from peer funds' flows, with the weights equal to the (own) funds' percentage portfolio share holding of each respective security (see equation (1) in the text for details). Panel B shows results for two variables constructed analogously but using buy and sell pressure separately to allow for asymmetries. To ease interpretation, these variables are expressed in standard deviation units. We report the baseline estimates in columns (1) and (4), results for subsamples of illiquid (high-yield) funds and in the crisis (columns (2) to (3) and (5) to (6), Panel A), for a specification that adds time (year) effects (columns (2) and (5), Panel B), and for a specification that adds fund covariates—fund and family size, expense ratios, and lagged performance (columns (3) and (6), Panel B). The time period is 1998 to 2014. All specifications include fund fixed effects. Standard errors are clustered by fund. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively. Variable definitions are provided in Section I of the [Internet Appendix](#).

Panel A: Baseline Analysis of Fund Performance						
	Monthly Return (pct)			Extreme Underperformance (pct)		
	All Funds (1)	Illiquid Funds (2)	Illiquid Times (3)	All Funds (4)	Illiquid Funds (5)	Illiquid Times (6)
Peer Flow Pressure $Sd_{i,t}$	0.35*** (0.01)	0.44*** (0.01)	1.22*** (0.03)	−2.52*** (0.07)	−3.00*** (0.11)	−6.69*** (0.27)
Fund Controls	No	No	No	No	No	No
FE	Fund	Fund	Fund	Fund	Fund	Fund
N	330,429	58,323	40,908	330,429	58,323	40,908
R ² (%)	1.95	4.21	8.79	6.16	2.41	13.99
Sd LHS	1.91	2.23	3.01	29.64	33.72	42.10
IQR LHS	1.58	1.99	2.82			
Panel B: Asymmetry and Robustness						
	Monthly Return (pct)			Extreme Underperformance (pct)		
	Asymmetry (1)	Add Time Controls (2)	Add Fund Controls (3)	Asymmetry (4)	Add Time Controls (5)	Add Fund Controls (6)
Peer Buy Pressure $Sd_{i,t}$	0.27*** (0.01)	0.15*** (0.01)	0.25*** (0.01)	−1.71*** (0.09)	−1.28*** (0.08)	−1.68*** (0.09)
Peer Sell Pressure $Sd_{i,t}$	−0.46*** (0.01)	−0.33*** (0.01)	−0.47*** (0.01)	4.28*** (0.11)	3.20*** (0.11)	4.45*** (0.11)
Fund Controls	No	No	Yes	No	No	Yes
FE	Fund	Fund, Time	Fund	Fund	Fund, Time	Fund
N	330,429	330,429	325,896	330,429	330,429	325,896
R ² (%)	3.03	10.11	3.45	7.59	12.29	8.12

Table V
Analysis of Fund Flows and Distress

This table reports regressions of monthly fund (net) flows (columns (1)–(3)) and extreme outflows (a dummy for net flows in the bottom decile of the distribution, columns (4)–(6)) on peer fund flows (net) pressure. In Panel A, the independent variable, *Peer Flow Pressure*, is a weighted sum of peers’ flow pressure, with weights calculated based on the asset allocation of a given fund. Specifically, we start with each security with information on holdings available from eMAXX and combine it with information on fund flows from CRSP to construct this measure for each fund-period by taking the weighted sum over the fund’s securities holdings of the security-by-security pressure from peer funds’ flows, with the weights equal to the (own) funds’ percentage portfolio share holding of each respective security (see equation (1) in the text for details). Panel B show results for two variables constructed analogously but using buy and sell pressure separately to allow for asymmetries. To ease interpretation, these variables are expressed in standard deviation units. We report the baselines estimates in columns (1) and (4), results for subsamples of illiquid (high-yield) funds and in the crisis (columns (2) to (3) and (5) to (6), Panel A), for a specification that adds time (year) effects (columns (2) and (5), Panel B), and for a specification that adds fund covariates—fund and family size, expense ratios, and lagged performance (columns (3) and (6), Panel B). The time period is 1998 to 2014. All specifications include fund fixed effects. Standard errors are clustered by fund. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively. Variable definitions are provided in Section I of the [Internet Appendix](#).

Panel A: Baseline Analysis of Fund Flows						
	Monthly Flows (pct)			Extreme Outflows (pct)		
	All Funds (1)	Illiquid Funds (2)	Illiquid Times (3)	All Funds (4)	Illiquid Funds (5)	Illiquid Times (6)
Peer Flow Pressure $Sd_{i,t}$	0.70*** (0.03)	0.99*** (0.11)	0.93*** (0.10)	−1.18*** (0.05)	−3.09*** (0.26)	−2.80*** (0.21)
Fund Controls	No	No	No	No	No	No
FE	Fund	Fund	Fund	Fund	Fund	Fund
N	330,429	58,323	40,908	330,429	58,323	40,908
R ² (%)	11.00	10.91	32.48	8.50	8.36	20.02
Sd LHS	8.55	9.51	9.16	30.51	32.74	33.78
IQR LHS	3.98	4.48	4.16			
Panel B: Asymmetry and Robustness						
	Monthly Flows (pct)			Extreme Outflows (pct)		
	Asymmetry (1)	Add Time Controls (2)	Add Fund Controls (3)	Asymmetry (4)	Add Time Controls (5)	Add Fund Controls (6)
Peer Buy Pressure $Sd_{i,t}$	0.30*** (0.03)	0.29*** (0.03)	0.22*** (0.03)	−0.71*** (0.07)	−0.72*** (0.07)	−0.73*** (0.07)
Peer Sell Pressure $Sd_{i,t}$	−0.92*** (0.03)	−0.67*** (0.03)	−0.59*** (0.03)	1.80*** (0.06)	1.75*** (0.07)	1.79*** (0.07)
Fund Controls	No	No	Yes	No	No	Yes
Clustering, FE	Fund	Fund, Time	Fund	Fund	Fund, Time	Fund
N	330,429	330,429	320,425	330,429	330,429	320,425
R ² (%)	11.66	13.76	14.44	8.92	9.18	9.33

by including lagged flows and other standard fund-level covariates (Panel B, columns (3) and (6)).

Finally, we examine the timing of the spillover effect using graphical analysis in Figure 1, Panels (b) and (c), and additional regression analysis in Table IA.I, Panel C. In line with the timing of the bond price reaction, the bulk of the spillover effect is concentrated in the immediate quarter and the following one, eventually reversing over the course of the third and fourth quarters. The dynamics of the effect on flows is somewhat more delayed and persistent.¹⁸

B.1 Economic Significance

Our estimates of the spillover effect on fund performance and flows are strongly economically significant but plausible. For example, the estimates imply that a 1-SD increase in *Peer Sell Pressure* leads to about 45 bp decrease in monthly returns, which is roughly equal to the sample mean of returns, and to an increase in the likelihood of extreme underperformance of over 4 pp. The spillover effect on fund flows is also economically significant, with a 1-SD increase in *Peer Sell Pressure* leading to about 90 bp decrease in monthly flows, which is roughly half the sample mean of flows, and to about 2 pp increase in the likelihood of extreme outflows. Our estimates of the price impact in Table III are also economically large, with a 1-SD increase in *Peer Sell Pressure* leading to about 1.7 pp drop in bond prices.

To further gauge the economic significance of the estimates and their plausibility, we conduct two exercises. First, we examine how a 1-SD change in our key explanatory variable moves a firm in the distribution of the outcomes.¹⁹ The estimated 45 (90) bp decrease in monthly returns (flows) in response to a 1-SD increase in *Peer Sell Pressure* corresponds to roughly one-quarter (one-tenth) of a standard deviation movement in the (conditional) distribution of fund returns (flows). The estimate for the price impact corresponds to about one-third of a standard deviation movement in the distribution of (log) bond

¹⁸ Table IA.VI summarizes additional analysis of funds' real and liquidity decisions to provide evidence on whether funds take actions in response to peer outflows, perhaps in an effort to mitigate the negative impact of such outflows. Specifically, we present results from estimating equation (2) for funds' real and liquidity decisions, where the primary outcomes are fund fees and cash holdings, respectively. The coefficient on *Peer Flow Pressure* is positive and highly statistically significant for both outcomes (columns (1) and (3)), indicating that funds tend to lower fees and deaccumulate cash possibly in an effort to attract investment and mitigate the harm of peer outflows on own flows. This result is consistent with recent evidence in Chernenko and Sunderam (2020) that exposure to fire sales is an important determinant of equity funds' liquidity policies. There is also evidence that funds are more likely to introduce rear-load fees in response to peer outflows, which is also consistent with an attempt to reduce harm (column (2)). These actions do not appear to undo the negative spillover effect, however, as there is a negative relation between *Peer Flow Pressure* and a common measure of the fund's asset illiquidity, the Roll index (column (4)).

¹⁹ Given the inclusion of firm fixed effects in the regression analysis that produces the point estimates, we use the within-firm distribution (i.e., the distribution after removing firm fixed effects) as a benchmark.

price changes. Thus, the relative magnitudes of our estimates at the different steps of the chain of events that leads to spillovers plausibly square together. Second, we compare the marginal effect of *Peer Sell Pressure* to that of standard fund-level covariates, such as fund and family size and fund fees, as well as lagged flows and performance. We calculate these marginal effects by multiplying the respective estimates by the within-fund standard deviation of each right-hand-side variable (see Table IA.IX). The marginal impact of *Peer Sell Pressure* for both performance and flows is of the same order of magnitude as any of these covariates, and is much larger than that of a change in fees, which further corroborates the notion that spillover effects are strong.

C. Second-Round Price Impact

If fire-sale spillovers lead to redemptions, as we show in Table V, an intriguing possibility is that redemptions may lead in turn to further sales and thus to a spillover effect on prices or a “second-round” price impact. Our final set of tests considers this fourth and last step of the spillover chain. In addition to closing the fire-sale feedback loop by tracing its effect back to prices, the analysis shows that fire-sale spillovers matter more broadly for debt markets.

To test whether peers’ flow-related trading generates feedbacks that are strong enough to further depress asset values, that is, whether fire-sale spillovers lead to a second-round price impact on debt securities, we employ a bond event study approach analogous to the initial analysis of the price impact. The results are summarized in Table VI, which reports regressions of monthly (changes in log) bond prices on a proxy for trades under peer fund (net) flow pressure. Here, the independent variable, *Peer Flow Pressure*, is meant to capture bonds for which large fractions of trading are accounted for by mutual funds experiencing significant peer outflow pressure. Accordingly, it is now defined as the bond-level counterpart of our fund-level spillover measure (see equation (1)), that is,

$$Peer\ Flow\ Pressure_{b,t} = \frac{Peer\ Flow\ Induced\ Buys_{b,t} - Peer\ Flow\ Induced\ Sales_{b,t}}{Offering\ Value_b},$$

$Peer\ Flow\ Induced\ Buys_{b,t} = \sum_j (\max(0, \Delta Holdings_{j,b,t}) | Peer\ Flow\ Pressure_{j,t} > Percentile(90th))$ and $Peer\ Flow\ Induced\ Sales_{b,t} = \sum_j (\max(0, -\Delta Holdings_{j,b,t}) | Peer\ Flow\ Pressure_{j,t} < Percentile(10th))$.

Panel A of Table VI reports the main results for a baseline specification that controls for bond (CUSIP) effects, with monthly (changes in log) bond prices (columns (1)–(3)) and crash (a dummy for monthly changes in log bond prices in the bottom decile of the distribution, columns (4)–(6)) as the outcome variables. The explanatory variables of interest are *Peer Flow Pressure* in Panel A and *Peer Buy Pressure* and *Peer Sell Pressure* in Panel B. The coefficient on *Peer Flow Pressure* is positive and highly statistically significant for both specifications, indicating that sales by mutual funds whose peers are experiencing outflow pressure tend to harm bond valuations. The result is stronger when illiquidity is higher (Panel A, columns (2), (3), (5), and (6)) and for peer sell

Table VI
Analysis of Second-Round Bond Price Impact

This table reports regressions of monthly (changes in log) bond prices (columns (1)–(3)) and price crash (a dummy for change in log prices in the bottom decile of the distribution, columns (4)–(6)) on a variable that proxies for trades under peer fund flow (net) pressure. Panel A shows results for a peer flow-pressure variable constructed as peer flow-related net buy pressure using fund flows as in Coval and Stafford (2007). Specifically, we start with each security with information on holdings available from eMAXX and combine it with information on fund flows from CRSP to construct first a fund-level measure that for each fund-period takes the weighted sum over the fund's securities holdings of the security-by-security pressure from peer funds' flows, with the weights equal to the (own) funds' percentage portfolio share holding of each respective security (see equation (1) in the text for details). Next, we combine this measure and fund trading to rank securities based on a security-level version of *Peer Flow Pressure*, which is constructed similarly to Coval and Stafford (2007) as the sum of the value of net "forced" fund buys (buys by funds with peer inflow pressure minus sells by funds with peer outflow pressure relative to the offering value of the bond), with "forced" buys and sales defined based on the top and bottom deciles of the distribution of the fund-level peer flow pressure, respectively. Panel B shows results for peer buy and peer sell pressure separately to allow for asymmetries. To ease interpretation, these variables are expressed in standard deviation units. We report results for the overall sample (columns (1) and (4)), for subsamples of illiquid bonds and in the crisis (columns (2) to (3) and (5) to (6), Panel A), for a specification with time (year) effects, and for a matched-sample estimator based on bond offering value, maturity, and lagged spread. The time period is 1998 to 2014. All specifications include controls for bond or time effects. Standard errors are clustered by bond. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively. Variable definitions are provided in Section I of the [Internet Appendix](#).

Panel A: Monthly % Bond Price Changes (pct)						
	Baseline			Crash		
	All Bonds (1)	Illiquid Bonds (2)	Illiquid Times (3)	All Bonds (4)	Illiquid Bonds (5)	Illiquid Times (6)
Peer Flow Pressure $Sd_{b,t}$	0.36*** (0.01)	0.63*** (0.03)	1.07*** (0.07)	−0.69*** (0.06)	−1.87*** (0.12)	−2.14*** (0.27)
FE	Bond	Bond	Bond	Bond	Bond	Bond
N obs	429,449	83,156	47,724	429,449	83,156	47,724
N Bonds	10,880	4,698	3,008	10,880	4,698	3,008
R ² (%)	5.42	8.61	9.03	8.61	11.73	13.49
Sd LHS	4.99	9.33	9.38	29.64	40.15	39.70
IQR LHS	1.93	3.68	3.33			
Panel B: Asymmetry & Robustness— Monthly % Bond Price Changes (pct)						
	Baseline			Crash		
	Asymmetry (1)	Time FE (2)	Matched Sample (3)	Asymmetry (4)	Time FE (5)	Matched Sample (6)
Peer Buy Pressure $Sd_{b,t}$	0.31*** (0.02)	0.15*** (0.02)	0.31*** (0.02)	0.16 (0.10)	−0.01 (0.10)	0.01 (0.11)
Peer Sell Pressure $Sd_{b,t}$	−0.29*** (0.01)	−0.24*** (0.01)	−0.29*** (0.01)	1.84*** (0.08)	1.16*** (0.08)	1.88*** (0.08)
FE	Bond	Time	Bond	Bond	Time	Bond
N obs	429,449	429,449	429,449	429,449	429,449	429,449
R ² (%)	5.62	4.61	5.62	9.25	10.36	9.26

pressure relative to peer buy pressure (Panel B, columns (1) and (4)) and it is robust to controlling for common shocks by including time effects (Panel B, columns (2) and (5)) and for omitted variables by matching on similar bonds based on bond offering value, bond maturity, and previous spread (Panel B, columns (3) and (6)). The spillover effect on bond prices is also economically significant and strongly asymmetric for crash likelihood, with a 1-SD increase in peer sell pressure leading to an almost 2 pp higher likelihood of a large bond price drop.²⁰ Taken together, these results indicate that fire-sale spillovers have material valuation consequences in the bond market.

III. Evidence from Three Experiments and Mechanism

The challenge in interpreting the baseline OLS estimates of the spillover effect is a manifestation of the “reflection problem” and boils down to a standard endogeneity problem: Flows are endogenously related to fund characteristics, so the challenge is to distinguish peer funds’ flow-driven trading from trading driven by changes in fund flows due to changes in fundamentals or “shocks” that are common across funds that hold the same securities. To address this concern, we use three quasi-experiments. Details on the regression specification are in Section II.C. Below, we discuss the treatment variables for each experiment. Section I of the [Internet Appendix](#) provides additional details.

The three peer treatment variables are as follows. First, we exploit plausibly independent time-series variation in peer flows, or peer flow “shocks,” due to the 2003 mutual fund trading scandal to implement a differences-in-differences (DD) design, which is similar to Anton and Polk (2014). The scandal was unexpected and involved 25 fund families settling allegations of illegal trading that included market timing and late trading with the then Attorney General of the state of New York, Elliot Spitzer. A well-replicated finding (see, for example, Kisin (2011) and McCabe (2009)) is that funds of implicated families experienced significant outflows that were long-lasting, arguably due to reputation effects. Important to convincingly rule out common shocks, fund families not implicated in the scandal did not experience direct contemporaneous shocks to their flows. In our first experiment, $Treatment_{j,t}$ is a dummy equal to one after a fund is involved in the mutual fund scandal of 2003 (“Spitzer 2003”) and zero otherwise.²¹

²⁰ [Internet Appendix](#) Table IA.I, Panel B, examines the timing of the price impact. The bulk of the effect is concentrated in the immediate quarter in which the sales happen and the following quarter (column (3)). The results in Table VI are robust to controlling for the first-round flow pressure variable, which leaves the estimates in all specifications little changed relative to those that are reported.

²¹ We obtain information on the identity and timing of the fund families involved from FAC-TIVA searches and from multiple sources that are standard in prior literature (primarily McCabe (2009) and references therein). After cross-checking the sources for consistency, we are able to include the following 18 treated fund families in our analysis: Alliance Bernstein 9/03, American Funds 12/03, Excelsior/Charles Schwab 11/03, Columbia Funds 1/04, DWS Investments 1/04, Federated 10/03, Franklin Templeton 2/04, Gabelli Funds 9/03, Invesco 11/03, Janus 9/03, Loomis

To further alleviate concerns about the exclusion restriction, we construct a second treatment based on an RD design that exploits sharp changes in peer flows around Morningstar five-star ratings. Each month, Morningstar issues mutual fund ratings based on arbitrary performance cutoffs of a risk-adjusted return, which are plausibly unrelated to changes in common industry fundamentals.²² Ratings are coarse and range from one star for worst funds to five stars for best funds. It is well documented that higher rated funds tend to enjoy inflows (see, for example, Del Guercio and Tkac (2008)). We exploit the coarse nature of the ratings and the fact that Morningstar makes its sorting variable, the risk-adjusted return, available to construct our treatment variable. For this second experiment, $Treatment_{j,t}$ is a dummy that takes a value of one for funds that are right below their respective rating category threshold and zero for funds that are right above it. To better isolate “quasi-random” variation around the thresholds, as is standard in RD designs, we only include funds that are close to their respective rating category threshold, which we measure as those funds whose Morningstar (annualized) risk-adjusted return is within 3 pp around the respective rating category threshold. This choice of bandwidth involves roughly one-quarter of the peer fund-month observations. The identifying assumption of this RD approach is that for funds that are “close” to their rating threshold, differences in ratings are close to a coin toss and, as such, constitute a plausible “quasi-random” treatment.

Because Morningstar ratings are coarse, there is a sharply nonlinear relation between flows and performance around the arbitrary performance thresholds that determine the rating assignment. The sharp nonlinearity provides for identification of the treatment effect under mild conditions. More technically, identification requires that the so-called local continuity assumption hold, which requires that all factors other than the treatment variable vary continuously around the rating thresholds. Intuitively, for the treatment effect β to not be identified, it must be the case that the unobserved component of the outcome variable, say fund i 's flows, $v_{i,t}$, exhibits an identical discontinuity around peer funds' performance as that defined by Morningstar rating thresholds. That is, even if $v_{i,t}$ is correlated with the distance between the peers' performance and their respective rating thresholds, our estimate of β is unbiased as long as $v_{i,t}$ does not exhibit precisely the same discontinuity around peer fund performance as *Peer Treatment Pressure* $_{i,t}$. To further isolate the discontinuity, in all the RD specifications we control for smooth (linear) functions of the distance from thresholds (see Roberts and Whited (2013) for a treatment of RD implementation with a focus on finance applications).

Sayles 11/03, Massachusetts Financial Services 12/03, PIMCO 2/04, Prudential Investments 11/03, Putnam 10/03, RS Investments 3/04, Sentinel 10/04, and Waddell & Reed 7/06.

²² Specifically, the arbitrary cutoffs are defined so that the distribution of funds within stars is an approximate bell curve: one (10%), two (22.5%), three (35%), four (22.5%), and five stars (10%). See Morningstar (2009) for additional details. Barrot (2016) also uses Morningstar ratings for identification.

We also consider a third “shift-share” treatment that exploits peers’ differential exposure to the collapse of the convertible bond market in 2005.²³ The collapse is documented in Mitchell, Pedersen, and Pulvino (2007), who show that funds specializing in convertible bonds experienced large redemptions starting in the first quarter of 2005 that were long lasting and led to fire sales and the eventual collapse of valuations in this asset class. An appealing feature of this event is that it was not anticipated and it originated from a specialized asset class. Similar to the mutual fund scandal, the fact that funds not specialized in convertible bonds did not experience direct contemporaneous shocks to their flows over this period helps address common shocks. We construct the $Treatment_{j,t}$ indicator variable for this additional experiment by interacting an indicator for the time period after the collapse of the convertible bond market (in 2005), the “shift” variable, with a dummy that is equal to one for funds whose Lipper asset class is convertible bonds, the “exposure” variable. By constructing the peer treatment as an interaction, we are not exploiting common variation across funds over time, but rather the differential behavior of convertible bond funds over time, an approach analogous to that commonly used in labor economics to analyze the impact of labor demand shocks (Bartik (1991), Blanchard and Katz (1992)).

In Panel A of Tables VII to IX, we repeat the bond event study (columns (1) and (2)), the analysis of fund performance (columns (3) and (4)) and flows (columns (5) and (6)), and the analysis of the second-round price impact (columns (7) and (8)) using *Peer Treatment Pressure* as the independent variable (see equation (3) for definitions). The three peer treatments are defined based on either fund-specific negative shocks to peer flows due to the 2003 scandal (Table VII) or a plausibly exogenous decrease in peer flows due to a close miss of their next Morningstar five-star rating for peer funds that are close to their rating category threshold (Table VIII), as per our RD approach, which restricts the sample of peers to those funds whose Morningstar risk-adjusted return is within a narrow range (3 pp) around the respective rating category threshold. Results for the additional experiment that exploits peer funds’ exposure to the collapse of the convertible bond market in 2005 are reported in Table IX (see Section II.C and Section I in the Internet Appendix for the definition of *Peer Treatment Pressure* for each treatment).²⁴ Omitted common factors do not appear to be driving the spillover effect, which continues to hold across fund performance, flows, and bond pricing for all of the three experiments.

In Panel B of Tables VII to IX, we estimate equation (2) in a two stage least squares-instrumental variable (2SLS-IV) setting using *Peer Treatment Pressure* as an instrument for *Peer Flow Pressure* in the first-stage regression.

²³ We thank Erik Stafford for suggesting this test.

²⁴ In Table IA.VII we present additional evidence on the collapse of the Asset-Backed Security (ABS) market in (the third quarter of) 2007. The intuition for this variable is based on anecdotal evidence from the financial crisis as well as the evidence in Chernenko, Hanson, and Sunderam (2014) that funds with relatively high holdings of private-label ABS experienced above average outflows from the second half of 2007.

Table VII
Experiment #1—The 2003 Mutual Fund Scandal

This table reports results on the fire-sale spillover effect of the 2003 mutual fund scandal for each of the four main outcomes, bond price impact (monthly (changes in log) bond prices, columns (1) to (2)), fund performance (columns (3) to (4)), fund flows (columns (5) to (6)), and second-round bond price impact (monthly (changes in log) bond prices, columns (7) to (8)). The independent variable for the bond price-impact analysis, *Treatment Pressure*, is defined analogously to Coval and Stafford (2007) as treatment-related net buy pressure, where treatment is a dummy that is equal to one after a fund is involved in the mutual fund scandal of 2003. The independent variable for the spillover effect analysis, *Peer Treatment Pressure*, is defined analogously to *Peer Flow Pressure* as a weighted sum of peers' treatment pressure, where treatment is a dummy that is equal to one after a fund is involved in the mutual fund scandal of 2003 at the fund or at the security level in columns (3) to (6) and (7) to (8), respectively (see equation (3) in the text for details). Panel A reports results for OLS regressions using these variables (the "reduced form"). Panel B reports 2SLS-IV estimates for specifications that use the respective treatment variables as instruments for *Flow Pressure* and *Peer Flow Pressure* in a first-stage regression. Table IA.IV shows first-stage diagnostics. To ease interpretation, the variables are expressed in standard deviation units. The time period is 2000 to 2006. All specifications include fund fixed effects. Standard errors are clustered by fund. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively. Variable definitions are provided in Section I of the Internet Appendix.

Panel A: Baseline

	Bond Price Impact		Fund Performance		Fund Flows		Second-Round Price Impact*	
	All (1)	Top-Bottom (2)	Ret (3)	Large Und. (4)	% Flows (5)	Large Outfl. (6)	All (7)	Top-Bottom (8)
Treatment Pressure $Sd_{b,t}$	2.06*** (0.41)	3.48*** (0.52)						
Peer Treatment Pressure $Sd_{i,t}$			0.26 *** (0.03)	-0.95*** (0.29)	0.77 *** (0.07)	-0.75*** (0.18)	0.31 *** (0.03)	0.29*** (0.04)
FE	Bond	Bond	Fund	Fund	Fund	Fund	Bond	Bond
N obs.	91,505	49,122	41,640	41,640	41,640	41,640	91,505	15,989
Sd LHS	4.99		1.91		8.55		4.99	
IQR LHS	1.93		1.58		3.98		1.93	

(Continued)

Table VII—Continued

Panel B: 2SLS-IV							
	Bond Price Impact		Fund Performance		Fund Flows		Second-Round Price Impact*
	All (1)	Top-Bottom (2)	Ret (3)	Large Und. (4)	% Flows (5)	Large Outfl. (6)	All (7) Top-Bottom (8)
Instr. Flow Pressure $Sd_{i,t}$	2.83** (1.08)	2.37*** (0.51)					
Instr. Peer Flow Pressure $Sd_{i,t}$			0.68*** (0.14)	-2.49*** (0.17)	1.19*** (0.12)	-0.97*** (0.24)	0.46*** (0.04)
FE	Bond 91,505	Bond 49,122	Fund 41,640	Fund 41,640	Fund 41,640	Fund 41,640	Bond 91,505
N obs.							Bond 15,989

Table VIII
Experiment #2—Morningstar Rating RD

This table reports results on the fire-sale spillover effect of Morningstar five-star ratings for each of the four main outcomes, bond price impact (monthly (changes in log) bond prices, columns (1) to (2)), fund performance (columns (3) to (4)), fund flows (columns (5) to (6)), and second-round bond price impact (monthly (changes in log) bond prices, columns (7) to (8)). The independent variable for the bond price-impact analysis, *Treatment Pressure*, is defined analogously to Coval and Stafford (2007) as treatment-related net buy pressure, where treatment is a dummy that is equal to one for funds that are just below their rating category threshold as defined by those funds whose Morningstar risk-adjusted return is within 3 pps of the respective rating category threshold. The independent variable for the spillover effect analysis, *Peer Treatment Pressure*, is defined analogously to *Peer Flow Pressure* as a weighted sum of peers' treatment pressure, where treatment is a dummy that is equal to one for funds that are just below their rating category threshold at the fund or at the security level in columns (3) to (6) and (7) to (8), respectively (see equation (3) in the text for details). Panel A reports results for OLS regressions using these variables (the "reduced form"). Panel B reports 2SLS-IV estimates for specifications that use the respective treatment variables as instruments for *Flow Pressure* and *Peer Flow Pressure* in a first-stage regression. Table IA.IV shows first-stage diagnostics. To ease interpretation, the variables are expressed in standard deviation units. The time period is 1998 to 2014. All specifications include fund fixed effects and control for a smooth (linear) function of the peer distance from thresholds. Standard errors are clustered by fund. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively. Variable definitions are provided in Section I of the Internet Appendix.

Panel A: Fund & Time Fixed Effects Estimators								
	Bond Price Impact		Fund Performance		Fund Flows		Second-Round Price Impact*	
	All (1)	Top-Bottom (2)	Ret (3)	Large Und. (4)	% Flows (5)	Large Outfl. (6)	All (7)	Top-Bottom (8)
Treatment Pressure $Sd_{b,t}$	0.45*** (0.02)	0.45*** (0.02)						
Peer Treatment Pressure $Sd_{i,t}$			0.33 *** (0.00)	-3.15*** (0.05)	0.19 *** (0.02)	-0.85*** (0.05)	0.44 *** (0.01)	0.46*** (0.02)
FE	Bond	Bond	Fund	Fund	Fund	Fund	Bond	Bond
N obs.	429,449	129,956	330,429	330,429	330,429	330,429	429,449	75,733
Sd LHS	4.99		1.91		8.55		4.99	
IQR LHS	1.93		1.58		3.98		1.93	

(Continued)

Table VIII—Continued

Panel B: 2SLS-IV							
	Bond Price Impact		Fund Performance		Fund Flows		Second-Round Price Impact*
	All (1)	Top-Bottom (2)	Ret (3)	Large Und. (4)	% Flows (5)	Large Outfl. (6)	All (7) Top-Bottom (8)
Instr. Flow Pressure $Sd_{i,t}$	5.87** (0.36)	5.13*** (0.41)					
Instr. Peer Flow Pressure $Sd_{i,t}$			1.17 *** (0.01)	-10.10*** (0.17)	1.05 *** (0.18)	-2.61 *** (0.21)	1.04 *** (0.03)
FE	Bond	Bond	Fund	Fund	Fund	Fund	Bond
N obs.	429,449	129,956	330,429	330,429	330,429	330,429	429,449 75,733

Table IX
Experiment #3—Collapse of the Convertible Bond Market

This table reports results on the fire-sale spillover effect of the collapse of the convertible bond market (see Mitchell, Pedersen, and Pulvino (2007)) for each of the four main outcomes, bond price impact (monthly (changes in log) bond prices, columns (1) to (2)), fund performance (columns (3) to (4)), fund flows (columns (5) to (6)), and second-round bond price impact (monthly (changes in log) bond prices, columns (7) to (8)). The independent variable for the bond price-impact analysis, *Treatment Pressure*, is defined analogously to Coval and Stafford (2007) as treatment-related net buy pressure, where treatment is a dummy that is equal to one after the market collapse (2005) for a fund whose Lipper class is convertible bonds and zero before the collapse (2003). The independent variable for the spillover effect analysis, *Peer Treatment Pressure*, is defined analogously to *Peer Flow Pressure* as a weighted sum of peers’ treatment pressure, where treatment is a dummy that is equal to one after the market collapse (2005) for a fund whose Lipper class is convertible bonds and zero before the collapse (2003) at the fund or at the security level in columns (3) to (6) and (7) to (8), respectively (see equation (3) in the text for details). Panel A reports results for OLS regressions using these variables (the “reduced form”). Panel B reports 2SLS-IV estimates for specifications that use the respective treatment variables as instruments for *Flow Pressure* and *Peer Flow Pressure* in a first-stage regression. Table IA.IV shows first-stage diagnostics. To ease interpretation, the variables are expressed in standard deviation units. The time period is 2003 (precollapse) versus 2005 (postcollapse). All specifications include fund fixed effects. Standard errors are clustered by fund. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively. Variable definitions are provided in Section I of the Internet Appendix.

Panel A: Fund & Time Fixed Effects Estimators							
	Bond Price Impact		Fund Performance		Fund Flows		Second-Round Price Impact*
	All (1)	Top-Bottom (2)	Ret (3)	Large Und. (4)	% Flows (5)	Large Outfl. (6)	All (7) Top-Bottom (8)
Treatment Pressure $Sd_{b,t}$	0.21*** (0.01)	0.18*** (0.02)	0.12*** (0.00)	-1.06*** (0.11)	0.11** (0.04)	-0.24*** (0.08)	0.14*** (0.02)
Peer Treatment Pressure $Sd_{i,t}$							0.10*** (0.02)
FE	Bond	Bond	Fund	Fund	Fund	Fund	Bond
N obs.	57,979	33,820	29,655	29,655	28,284	28,284	57,979
Sd LHS	4.03		1.53		9.92		5.08
IQR LHS	2.04		1.58		4.22		1.99

(Continued)

Table IX—Continued

Panel B: 2SLS-IV							
	Bond Price Impact		Fund Performance		Fund Flows		Second-Round Price Impact*
	All (1)	Top-Bottom (2)	Ret (3)	Large Und. (4)	% Flows (5)	Large Outfl. (6)	All (7) Top-Bottom (8)
Instr. Flow Pressure $Sd_{i,t}$	3.43** (0.50)	2.89*** (0.61)					
Instr. Peer Flow Pressure $Sd_{i,t}$			0.46 *** (0.03)	−4.12*** (0.42)	0.57 *** (0.22)	−1.09*** (0.36)	0.76 *** (0.05)
FE	Bond	Bond	Fund	Fund	Fund	Fund	Bond
N obs.	57,979	33,820	29,655	29,655	28,284	28,284	57,918 24,115

Robustly across the three treatments, the IV estimates, reported in Panel B of the respective tables, remain strongly significant and are generally of the same order of magnitude and somewhat larger than their OLS counterparts in Tables III to VI (Panel A, column (1)), consistent with the spillover effect from unanticipated flows being larger than that from anticipated flows and omitted common shocks leading, if anything, to downward bias in our ordinary least squares (OLS) estimates. The estimated elasticities are large but plausible across outcomes. For example, a 1-SD change in instrumented *Peer Flow Pressure* leads to about 1 pp change in fund returns and 1 pp change in flows for the specification that uses the RD treatment.

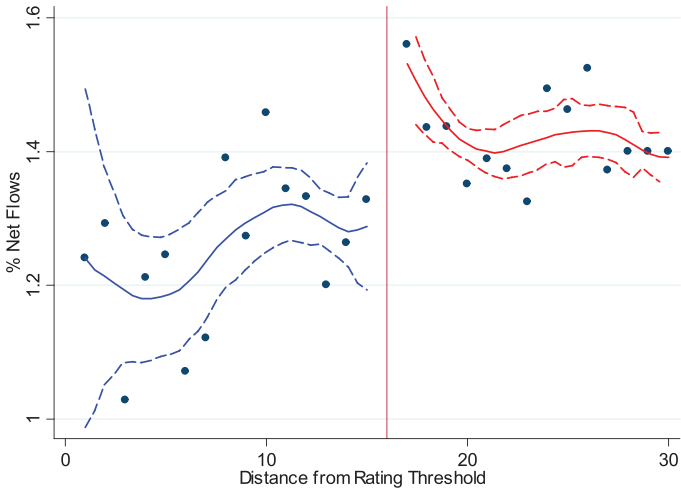
A. Identification Assumptions, Falsification, and Additional Sensitivity Tests

The key identifying assumption for our treatments to be valid is that they are driven by events that are relevant for peer fund flows (the so-called “relevance condition”), but that are otherwise plausibly random with respect to fund *i*’s flows and performance (the so-called “exclusion restriction condition”). More specifically, the identification assumption behind the RD strategy is the local continuity assumption, which requires that all factors other than the treatment variable vary continuously at the rating thresholds. For the DD strategies, the identifying assumption is that there are no differential trends pretreatment between treated versus untreated funds. If these assumptions hold, then the estimates of β in equation (2) can be interpreted as causal.

To alleviate concerns about the internal validity of our estimates, we implement a battery of diagnostic tests, which include a first-stage validity test as well as a balancing test on the covariates. Panels A and B of Table IA.IV report first-stage diagnostics that confirm the validity of our three treatments. In line with the prior literature, we confirm that funds experience significant flow-driven pressure subsequent to becoming implicated in the 2003 scandal (column (1)), after close misses of Morningstar ratings thresholds (column (2)), and after the collapse of the convertible bond market. The first-stage estimates are all large and strongly statistically significant with good explanatory power (high R^2), which supports the relevance condition. In a standard test of pretrends, we do not detect significant differences in bond valuations based on the three treatment variables pretreatment (Panel C). We use balancing tests to further assess whether funds are otherwise similar in terms of observables. For example, if funds have a discontinuity in characteristics around the rating thresholds, that would constitute a violation of the local continuity assumption. Panel D of Table IA.IV reports estimates of our baseline specification with observable predetermined fund characteristics as outcome variables, none of which is statistically significant.

Figure 2 illustrates and supports the internal validity of our RD approach. Graphical analysis of average fund flows reveals that there is a sharp discontinuity in the relation between funds and performance (the risk-adjusted return) around the Morningstar rating performance thresholds, with the discontinuity especially pronounced for funds that are in the top (4 or 5) and bottom (1 or 2)

Panel (a): "Close" top and bottom (1 or 2 and 4 or 5) Morningstar Ratings



Panel (b): "Close" intermediate (2 or 3 and 3 or 4) Morningstar Ratings

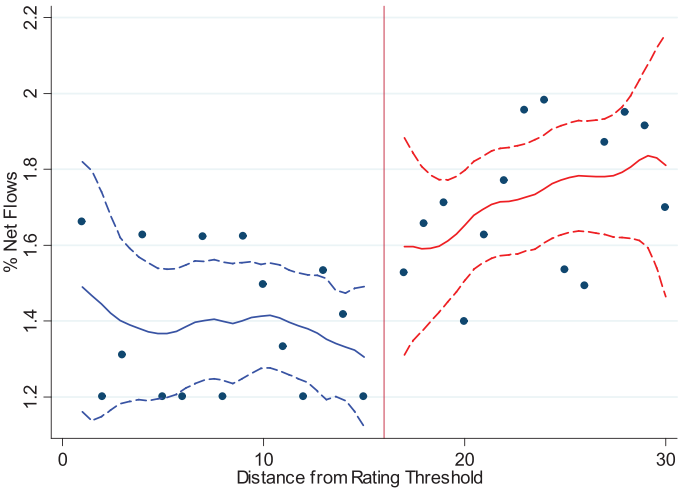


Figure 2. "Close" Morningstar five-star ratings and fund flows. The starting sample consists of the funds whose Morningstar five-star rating is close to their respective rating category threshold, as defined by those funds whose Morningstar risk-adjusted return is within 3 pps around the respective rating category threshold. The time period is 1992 to 2014. This figure plots the average of monthly (net) fund flows (vertical axis) against the forcing variable, the distance between the risk-adjusted return and its respective rating category threshold (horizontal axis). Panel (a) restricts the sample to top and bottom ratings only (1 or 2, and 4 or 5), while Panel (b) is for "close" intermediate ratings (2 or 3, and 3 or 4). Observations to the left of the red line correspond to "close" misses, that is, funds that are just below the threshold. Each circle is the average monthly (net) fund flows within the derived bin width, with each bin containing multiple underlying observations. The plotted bin bandwidth is for 30 equidistant bins, which correspond to the range of the x-axis. The plotted bin are fitted values from polynomial regressions on either side of the discontinuity. Standard errors are calculated via bootstrapping and the dashed lines represent the upper and lower 95% confidence intervals. (Color figure can be viewed at wileyonlinelibrary.com)

rating bins (Panel A). Table [IA.V](#) reports additional specification checks on the RD tests. Panel A of Table [IA.V](#) allows for heterogeneity of the RD estimates by each of the four Morningstar rating category thresholds. While the estimates remain generally significant across thresholds, in line with Figure [2](#), we find evidence of heterogeneity, with the estimates for the top and especially the bottom thresholds driving the overall result. We also implement additional tests to examine the sensitivity of our RD estimates to alternative specifications, which include using higher order polynomials of the distance from the rating threshold (Panels B and C of Table [IA.V](#)) and using a narrower bandwidth for the discontinuity sample (Panel D of Table [IA.V](#)).²⁵ Estimates remain robustly close to our baseline across all of these alternatives. These findings lend further support to the notion that our baseline peer effects represent valid RD estimates.

Finally, we conduct two falsification tests to further increase confidence about the exclusion restriction. The first is a test of pretrends for the fund-level DD analysis. Specifically, we estimate our baseline specification using the one-quarter-ahead lead of *Peer Treatment Pressure*, both for the 2003 mutual fund scandal and the 2005 collapse of the convertible bond market. If the scandal or the market collapse were anticipated or reflected omitted fund characteristics, one might expect an “effect” of the peer treatment even prior to these events. The results in Table [IA.VIII](#), Panel B, indicate that this is not the case, as the estimates are small and statistically insignificant for both placebo treatments (columns (1), (2), (5), and (6)).

The second falsification test corroborates the exclusion restriction for the RD analysis and follows an implementation recommended by Imbens and Lemieux (2008). We define the placebo dummy as the median of the forcing variable in the two subsamples on either side of the threshold. To that end, we take funds whose Morningstar risk-adjusted return is within 3 pps around the respective rating category threshold and, for each subsample just-above or just-below the rating category thresholds, we replace the treatment dummy with a placebo treatment dummy based on whether the distance from the threshold is above or below the within-subsample median. The results in columns (3) and (4) of Table [IA.VIII](#) show that the jump in peer returns and flows occurs only at the true rating threshold.²⁶

²⁵ Table [IA.V](#), Panel E, summarizes the results of an additional specification check. We show that the statistical significance of the baseline RD estimates is little changed when we use an alternative double-clustering of the standard errors that adds clustering by time. We report results for double-clustering on month, but we experimented with double-clustering by lower frequency time variables, namely, quarter and year, both of which leave the standard errors little changed relative to the baseline.

²⁶ To offer further reassurance about the exclusion restriction, in Panels B and C of Table [IA.X](#) we examine the sensitivity of the estimates to adding fund and time controls. The effects for each of the three treatments remain stable and strongly statistically and economically significant even after controlling for potential differences in the composition of treated funds (that are time-varying and thus may not be captured by fund effects) as well as for aggregate shocks with the inclusion of time effect to further address concerns about common shocks.

Taken together, both graphical evidence and results of the more formal diagnostic and falsification tests support our conclusion that the peer treatment effects represent valid estimates and hence bolster a causal interpretation of our results.

B. Mechanism

The economic mechanism that makes fire sales costly and leads to spillovers is downward-sloping demand for corporate bonds—it is through this mechanism that fire sales of a fund's portfolio securities, that is, sales that are forced by redemptions, have a price impact. The literature identifies several factors that can potentially lead to downward-sloping demand, including illiquidity due to transaction costs and, more broadly, slow-moving capital factors that make high-valuation bidders relatively scarce and lead to arbitrage persistence. A number of recent papers put forward potential explanations for the existence of persistent mispricing in financial markets. Mitchell, Pedersen, and Pulvino (2007) and Duffie (2010) discuss the role that slow-moving capital may play in allowing arbitrage opportunities to exist for extended periods of time.²⁷ To the extent that purchasing a corporate bond requires the use of significant capital, any combination of these factors that sideline arbitrage capital can plausibly lead to downward-sloping demand for bonds.

By depressing security prices, flow-related sales lead to spillovers because the valuation losses hurt the performance of peer funds that hold the same securities. In turn, spillovers may lead to redemptions at peer funds through the performance-flow relationship. The evidence on fund performance and flows that we document in Tables IV and V is in line with this mechanism. In Internet Appendix Tables IA.II and IA.III we present tests to further corroborate the mechanism. Specifically, to increase confidence that the source of the spillover is indeed bond price changes, we rerun our baseline fund-level regressions for returns and flows using as the main explanatory variable exposure to the percentage price changes of the funds' bond holdings (the “reduced form,” Table IA.II) or, in a 2SLS-IV setting where we instrument for exposure to price changes, using *Peer Flow Pressure* and then including the price changes as predicted by peer flow pressure in the second-stage tests (Table IA.III, Panels A and B). Estimates of the reduced form are strongly statistically significant and large, confirming that fund performance and flows are quite sensitive to the value of asset holdings. Importantly, estimates for instrumented bond price changes are also large, which helps increase confidence that fire-sale spillovers affect performance and flows in an instrumental variables sense via their impact on the value of securities.

²⁷ Shleifer and Vishny (1997), Gromb and Vayanos (2002), Liu and Longstaff (2005), Fostel and Geanakoplos (2008), Gorton and Metrick (2009), and Ashcraft, Garleanu, and Pederson (2010) argue that margins, haircuts, and other collateral-related frictions may allow arbitrage or deviations from the law of one price to occur. Brunnermeier and Pedersen (2009) emphasize the role that the availability of funding may play in liquidity effects on security prices.

Finally, in Panel C of Table IA.III we use 2SLS-IV to clarify the flow-performance relation in the spillover to fund flows. We replicate the main analysis of flows using performance instrumented with peer flow pressure as the main explanatory variable. The 2SLS estimates are remarkably close to their reduced-form counterpart presented in the main analysis (Table V), which supports the notion that the spillover to flows is via fund performance. While the two sets of 2SLS estimates provide additional support for the price-impact mechanism, they do not rule out the possibility that other economic mechanisms may also contribute to the spillover. One such mechanism is information and investor learning, whereby fire sales lead to reputation spillovers. For example, it may be the case that severe underperformance of a high-yield fund leads investors to punish other high-yield funds.

Disentangling the relative contribution of different mechanisms is beyond the scope of this paper and constitutes a promising venue for future work. However, in Panel A of Table IA.VIII we provide additional support for the price-impact mechanism using placebo tests.²⁸ We consider two placebo variables, both of which are constructed as a weighted sum of peers' placebo flow pressure. In Panel A, columns (1) and (2) *Placebo Peer Flow Pressure* is constructed using weights calculated based on the asset allocation of a given fund but placebo flow pressure, which is defined using nonforced trades. Similar to Khan, Kogan, and Serafeim (2012), who interpret nonforced trades as more likely to be information-driven trades and define them as *Not – Flow Induced Trades* $_{b,t}^{j \neq i} = \sum_{j \neq i} (\Delta Holdings_{j,b,t} | \text{Percentile}(10th) < Flows_{j,t} < \text{Percentile}(90th))$. In Panel A, columns (3) and (4) *Placebo Peer Flow Pressure* is constructed using forced sales but placebo weights, which are calculated based on the past asset allocation (over the previous year) of a fund that remains active in a given Lipper asset class but ceases to hold a given security in a given month. The results indicate that there is no spillover for asset sales that are not flow-driven or for funds that operate in the same markets but do not have common holdings, both of which corroborate the price-impact mechanism.

IV. Implications for Financial Stability

In this section, we examine the financial stability implications of fire-sale spillovers. The term “financial stability” is broad and has been used in the literature for different kinds of market vulnerabilities. We examine whether fire-sale spillovers aggravate a specific type of market instability—volatility—by amplifying the effect of an initial shock to fund flows that is otherwise unrelated to fundamental asset values. The idea is that outflows at peer funds lead to a second round of outflows that further depress bond prices over and above the initial effect of an adverse shock. As a result, spillovers lead to higher volatility by increasing the exposure of funds and bonds to nonfundamental risk. To clarify the conditions and circumstances under which fire-

²⁸ We thank Marco di Maggio for this suggestion.

sale spillovers lead to volatility, we construct measures of fund exposure to spillovers based on our estimates and explore the relation between these measures and the volatility of fund and bond returns. We also use a calibrated structural model to gauge the quantitative importance of the mechanism.

We construct two measures of exposure to fire-sale spillovers, namely, systemicness and vulnerability, which we use in regression analysis of bond and fund return volatility similar to Greenwood and Thesmar (2011). For ease of exposition, we calculate the measures annually and at the fund family level. Recall that *Peer Flow Pressure* for fund i is defined as a weighted sum of other funds' flow pressure, with weights equal to the pair-wise coinvestment of funds' asset allocations across individual asset holdings, $w_{i,j}$. Using the pair-wise version of these weights, $W_{i,t}^j = \sum_b I_{b,t}^j * w_{b,t-1}$, for all $j \neq i$, where $I_{b,t}^j$ is an indicator variable equal to one for securities held by fund j at time t ,²⁹ and an estimate of the spillover effect, $\hat{\beta}$, we construct matrix B of order $n \times n = 254 \times 254$, which is defined in Section II of the [Internet Appendix](#). Each element of B, b_{ij} , gives the effect of a 100 bp increase in fund family j 's pressure on i 's current flows. Given B and an assumption for the attenuation factor, a , we calculate fund i 's vulnerability and systemicness, respectively, as $Vulnerability_i = \frac{1}{n} \sum_{s=1}^{\infty} \sum_{j=1}^n a^s b_{ij}^s$ and $Systemicness_i = \frac{1}{n} \sum_{s=1}^{\infty} \sum_{j=1}^n a^s b_{ji}^s$. The vulnerability measure captures own-fund exposure to peer fire sales and ranks as more vulnerable those funds for which peer outflows are more likely to lead to own outflows. The systemicness measures captures the degree of other funds' exposure to a fund's own fire sales and ranks as more systemic those funds whose outflows are more likely to lead to peer outflows. In general, these measures are higher when (i) there is more overlap in bond holdings with other funds, that is, when the pair-wise weights in $W_{i,t}^j$ are higher; and (ii) debt market conditions are such that forced sales have a larger price impact, that is, when the estimates of the spillover effect, $\hat{\beta}$, are higher.³⁰ In the remainder of this section we use regression analysis and a model-based calibration to illustrate these mechanisms.

Table X, Panel A, provides a descriptive analysis of the main cross-sectional correlates of the two measures. Systemic (vulnerable) funds are larger and more (less) diversified. Even then, standard fund characteristics have relatively low explanatory power especially for vulnerability, suggesting that, while systemic and vulnerable funds share some characteristics in common, our exposure measures do not simply reflect standard fund characteristics. To further develop intuition for our measures, consider the difference with concentrated holdings. A fund with relatively undiversified holdings would certainly be more exposed to bond-specific risks than a more diversified fund if it were forced to sell in response to outflows. However, holding concentration

²⁹ For example, for two funds that have the following holdings of a Bond C, Fund A holdings of Bond C = \$100 and Fund B holdings of bond C = \$150, the contribution of Bond C to the numerator of the pair-wise weight is \$100.

³⁰ Table IA.XI lists fund families that are ranked as most systemic (Panel a) and most vulnerable (Panel B) for the most recent year (2014).

Table X
Financial Stability—Implications for Individual Funds

This table presents fund-level financial stability implications of fire-sale spillovers. Panel A reports results of annual cross-sectional regressions of the systemicness and vulnerability scores on fund family characteristics. Panel B examines predictability regressions of fund return volatility, measured as the one-quarter- or one-year-ahead standard deviation of fund returns (σ_{it+1}), which is regressed on the vulnerability score. We also show results for a specification where the score is interacted with a *Crisis* indicator. Panel C examines the implications for fund return comovement with market-wide factors, which is measured as the one-quarter-ahead correlation between fund returns and the Vanguard Total Bond Market Index Fund return ($\beta_{it+1}^{Bond\ Mkt}$) in column (1) and the CRSP value-weighted market return ($\beta_{it+1}^{Bond\ Mkt}$) in column (2). Details on the definitions of the scores are in Section V and in Section II of the [Internet Appendix](#). All other variable definitions are provided in Section I of the [Internet Appendix](#).

Panel A: Correlates of Systemicness and Vulnerability		
	Fund Systemicness (1)	Fund Vulnerability (2)
Fund Size $_{i,t}$	0.12*** (0.11)	0.07*** (0.01)
Fund Holding Concentration (HHI) $_{i,t}$	-0.46 *** (0.17)	0.62*** (0.17)
Fund Performance $_{i,t}$	-0.58 (1.22)	-1.25 (1.27)
Expense Ratio $_{i,t}$	0.04 (0.05)	0.06 (0.05)
Turnover Ratio $_{i,t}$	0.05*** (0.02)	0.06*** (0.02)
Corporate Bond Holdings $_{i,t}$	0.12 (0.09)	0.24*** (0.09)
Foreign Bond Holdings $_{i,t}$	0.41*** (0.13)	0.42*** (0.13)
Government Bond Holdings $_{i,t}$	0.08 (0.12)	-0.22* (0.13)
N	2,237	2,237
R ²	8.07	4.18
Panel B: Implications for the Volatility of Fund Returns (pp)		
N = 326,575	Quarterly Vol.	Annual Vol.
[1] Vulnerability $_{i,t}$	0.44*** (0.01)	0.57*** (0.01)
R ²	0.77	1.73
[2] Vulnerability $_{i,t}$ *Crisis	2.36 *** (0.06)	2.14*** (0.05)
R ²	10.70	14.50
Panel C: Implications for the Comovement of Fund Returns (pp)		
N = 326,575	Quarterly β_{MKT}^{Bond}	Quarterly β_{MKT}^{Stock}
[3] Vulnerability $_{i,t}$	6.76*** (0.15)	8.25*** (0.10)
R ²	0.92	1.89

primarily captures exposure to fund-specific risk, that is, how exposed a fund portfolio is to fund- or bond-specific shocks. By contrast, the overlap of bond holdings across funds captures the extent to which shocks to any particular bond or asset class will be borne across a wide swath of funds, that is, the strength of transmission of fund- or bond-specific shocks to other funds. For example, if funds A and B hold the same asset and an exogenous shock forces A to liquidate the asset, the price of the asset will decline and therefore change the value of B's portfolio potentially leading B to also sell the asset at an unfavorable price. Thus, overlapping holdings are a potentially useful gauge of the strength of transmission of fire sales across funds.

In Table X, Panels B and C, we use regression analysis to test at the fund level our prediction that exposure to peer fire sales should forecast volatility. We use the one-quarter- or one-year-ahead monthly volatility of fund returns as the dependent variable and regress it on the fund vulnerability measure. As expected, the vulnerability measure has predictive power and is positively correlated with future volatility. For example, a 1-SD increase in fund vulnerability leads to an increase in annual volatility of about 0.6%, which is approximately one-third of the sample mean. The estimates are twice as large in the crisis, suggesting that spillovers may also contribute to procyclicality. Panel C examines this possibility more directly by testing whether the vulnerability measure also has forecasting power for peer-induced comovement with market-wide factors. We use one-quarter-ahead market β s of monthly fund returns with respect to an index of bond fund market returns as well as an index of stock market returns as the dependent variables and regress them on the vulnerability measure. As expected, the vulnerability measure has predictive power and is positively correlated with future β s, which further corroborates the notion that fire-sale spillovers are a source of nonfundamental risk.

Next, we examine sector- and market-wide implications. Table XI, Panels A and C, repeats the forecasting regression analysis of volatility for the aggregate bond mutual fund sector and for bond returns, respectively, for the systemicness and vulnerability scores. In Panel A, we construct value-weighted indices of fund returns and use one-quarter- or one-year-ahead monthly volatilities of the value-weighted returns as the dependent variable, which we regress on value-weighted indices of each of the two exposure measures. In Panel C, we regress the volatility of bond returns on measures of a bond's exposure to systemic and vulnerable funds, which are calculated by taking (lagged) holding-weighted averages of the systemicness and vulnerability scores of its holders. In the aggregate and at the bond level, both measures have predictive power and are positively correlated with future volatility, indicating that when the mutual fund sector as a whole or a bond on average becomes more exposed to spillovers—either because funds are more systemic and thus matter more

Table XI
Financial Stability—Implications for the Mutual Fund Sector and the Bond Market

This table reports sector-wide and bond market-level financial stability implications of fire-sale spillovers. Panel A and column (1) of Panel C repeat the volatility analysis of Table X (Panel B) for aggregate sector-wide returns and for bond returns, respectively, with Newey-West standard errors reported in Panel A. Column (2) of Panel C repeats the comovement analysis of Table X (Panel C) for bond returns. Panel B reports the results of a stress-test exercise that consists of calculating an aggregate counterfactual for the effect on sector-wide flows of flow-induced fire sales by the most systemic fund families. We report results for aggregate flows relative to aggregate assets under management and for several alternative adverse scenarios that vary by the size of the shock (5% versus 10% versus 20% reduction in portfolio holdings) as well as by which funds are shocked (most versus least systemic funds). Details on the definitions of the scores are in Section V and in Section II of the [Internet Appendix](#), which also has details on the implementation of the stress-test exercise. All other variable definitions are provided in Section I of the [Internet Appendix](#).

Panel A: Implications for the Volatility of Mutual Fund Sector Returns (pp)			
N = 192	Quarterly Vol. (1)	Annual Vol. (2)	
[1] Systemicness _t	0.99*** (0.40)	1.03*** (0.24)	
R ²	1.60	6.03	
[2] Vulnerability _t	1.03** (0.40)	1.06*** (0.24)	
R ²	2.20	6.87	
[3] Systemicness _t *Crisis	1.85*** (0.66)	1.81*** (0.27)	
R ²	18.89	45.20	
[4] Vulnerability _t *Crisis	1.83*** (0.66)	1.79*** (0.27)	
R ²	18.93	47.89	
Panel B: The Effect of Fire-Sales at Top 10 Systemic Funds – A Stress-Test Exercise			
Fire-Sale Shock Size =	5%	10%	20%
Jan–Dec 2014	−9.84%	−19.68%	−39.37%
Jan–Dec 2013	−10.86%	−21.72%	−43.45%
Jan–Dec 2009	−10.52%	−21.03%	−42.06%
Jan–Dec 2010	−10.34%	−20.68%	−41.35%
Jan–Dec 2011	−10.32%	−20.63%	−41.26%
Jan–Dec 2012	−10.25%	−20.49%	−40.98%
Jan–Dec 2008	−9.69%	−19.37%	−38.74%
Jan–Dec 1999	−4.83%	−9.67%	−19.34%
Bottom 10 Systemic Funds, Jan-Dec 2014	−0.01%	−0.02%	−0.04%
Bottom 100 Systemic funds, Jan-Dec 2014	−2.03%	−4.05%	−8.09%
Panel C: Implications for the Volatility & Comovement of Bond Returns (pp)			
N = 423,668	Quarterly Vol.	Quarterly β_{MKT}^{Bond}	
[5] Systemicness _{b,t}	1.74*** (0.03)	2.78*** (0.22)	
R ²	14.90	10.43	
[6] Vulnerability _{b,t}	1.25*** (0.03)	2.22*** (0.21)	
R ²	14.60	10.39	

for their peers, or because they are more vulnerable and thus more exposed to their peers—that leads to higher subsequent volatility.³¹

Table XI, Panel B, reports results of a stress-test exercise that calculates the effect of fire sales by the most systemic funds on sector-wide flows. We report results for aggregate flows as a percentage of aggregate assets under management and for several alternative adverse scenarios that vary by the size of the shock to the 10 most systemic funds (5% versus 10% versus 20% reduction in portfolio holdings). To illustrate how the effect changes over time, we report estimates for the most recent data (January to December 2014) as well as for the years in which spillovers had the greatest potential to lead to a systemic effect on mutual fund sector flows. For reference, we also report the estimates for the earlier data (1999) and for the case in which the 10 or even 100 least systemic funds are shocked. Details on the calculations are in Section II of the Internet Appendix. Shocks to the most systemic funds generally have large effects on the industry. In fact, it takes roughly the bottom half of the sector in terms of systemicness to be affected to trigger sector-wide outflows that are comparable in magnitude to those induced by a shock to just the 10 most systemic funds. The aggregate effects have become more severe in the crisis and postcrisis years, indicating that the degree of holding overlap has increased in recent years as the overall sector has dramatically increased its participation in the bond market.

Finally, we provide a quantitative assessment of the implications for volatility using a calibrated discrete-choice model of fund-flow decisions in the style of Hortacsu and Syverson (2004). Additional details are in Section II of the Internet Appendix, but the two key features are that there is a flow-performance relation, because investors in the model decide which fund to invest in based on an array of fund characteristics such as fund returns, and there are cross-fund fire-sale spillovers, because any given fund's return is positively (negatively) affected by the inflows (outflows) of its peers. Specifically, we assume that investors' utility from investing in fund i , u_{it} , depends on fund i 's return R_{it} and an array of standard fund characteristics X_{it} , which include fund size and the expense ratio,

$$u_{it} = \alpha \cdot R_{it} + X'_{it}\beta + \epsilon_{it}, \quad (4)$$

and that fund i 's return R_{it} is an increasing function of its peer funds' flows,

$$R_{it} = \gamma \cdot \left(\sum_{j \neq i} s_{jt} \cdot \omega_{jt} \right) + \zeta_{it}, \quad (5)$$

where s_{jt} is peer fund flow shares, ω_{jt} is the degree of portfolio overlap across funds, and ζ_{it} is the fund-specific component of returns. While the model is

³¹ Note that the systemicness score is included in the aggregate analysis (Table XI) but not in the fund-level analysis (Table X) because, while systemic funds should make their peers unambiguously riskier, it is less clear a priori whether they should be themselves riskier.

structural in the sense that we derive the equilibrium solution for fund flow shares and returns for a given set of parameter values, the specification for fund returns captures in reduced form the main mechanism through which fire-sale spillovers operate, which is that fund performance is hurt by outflows at peer funds.

The model is parsimonious enough to allow for a calibration that uses as inputs four main estimates from the data: the flow-performance relation, $\hat{\alpha}$, the relation between flows and standard fund characteristics, $\hat{\beta}$, the intensity of the spillover effect, $\hat{\gamma}$, and the time series of monthly fund-specific return shocks, $\hat{\zeta}_{it}$. Specifically, under standard assumptions on ϵ_{it} , the model can be solved as a fixed-point problem with two equations in two unknowns—fund flow shares, s_{it} , and fund performance, R_{it} —for each fund i . For a given set of parameters and for each fund, we determine the solution to the fixed-point problem using different starting points for s_{it} and R_{it} on the $(0, 1) \times (0, 1)$ grid and bootstrapped residuals. To calibrate the model, we estimate the parameters α and β from the data using equation (4) and data on fund flows and returns, which is analogous to estimating a standard flow-performance equation. We also estimate γ and ζ_{it} from the data using equation (5) and data on fund returns and peer fund flows. Finally, we calculate the model-implied equilibrium solutions for s_{it} and R_{it} at each point in time t and their volatility (standard deviation) for the estimated $\hat{\alpha}$, $\hat{\beta}$, $\hat{\gamma}$, and bootstrapped residuals $\hat{\zeta}_{it}$.

Figure 3 shows the results of the model calibration for the volatility of fund returns (Panel a) and flows (Panel b). We plot the average model-implied volatilities (standard deviations) for four different estimates of the spillover parameter, $\hat{\gamma}$, namely, 0.04 in the precrisis period (1998–2006), 0.1 in the postcrisis period (2010–2014), 0.2 for the first crisis subperiod (2007Q4 to 2008H1), and 0.3 for the second crisis subperiod (2008H2 to 2009H1), as well as for a counterfactual value of 0.4 that is even larger than those estimated in the crisis. We quantify the contribution of the spillover to volatility by plotting the model-implied volatilities under these different estimates of $\hat{\gamma}$ relative to the counterfactual benchmark of no spillover, that is, the model-implied volatilities for $\hat{\gamma} = 0$, which, for reference, implies a baseline standard deviation of returns (flows) of about 1.4% (10.4%). In the precrisis period, the contribution of spillovers to volatility was negligible, as the standard deviation of returns (flows) increased by just about 5% (3%) relative to the no-spillover benchmark. By contrast, the effect was very large especially in the second part of the crisis period, with the standard deviation of returns (flows) increasing by roughly one and a half times (24%), and it remained significant postcrisis, leading to an increase in the standard deviation of returns (flows) of about 22% (7%). Interestingly, the effect is outsized and potentially disruptive in the “deep-crisis” scenario, pointing to a nonlinear relation between volatilities and the intensity of the spillover. Overall, the model-based calibration indicates that spillovers had a large effect on volatility in the crisis and that the effect has abated but remained nonnegligible in recent years.

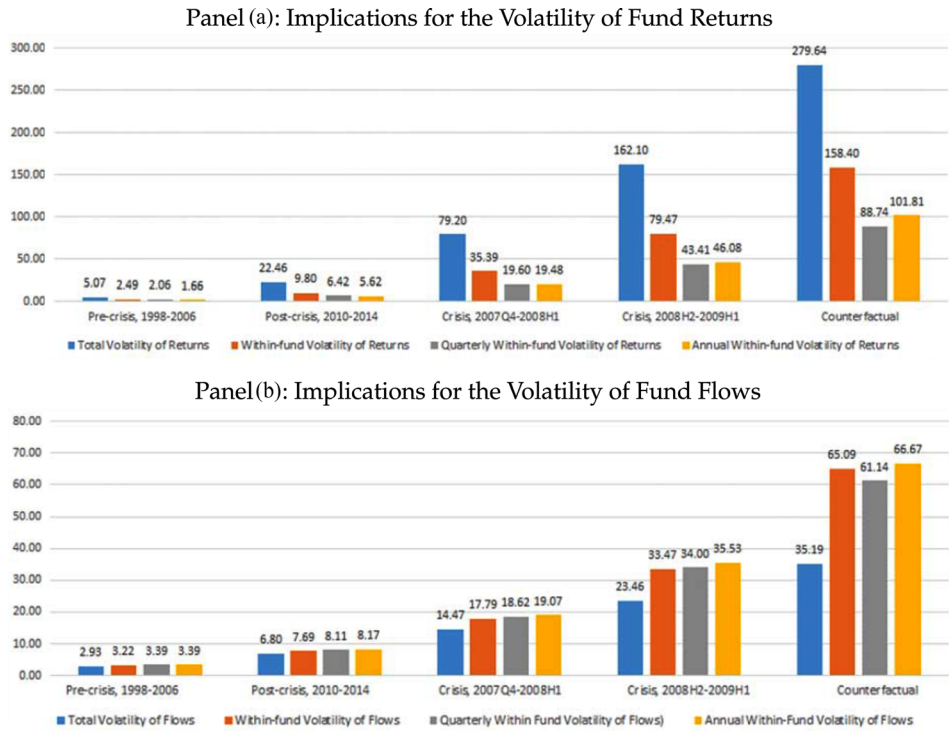


Figure 3. The financial stability implications of fire-sale spillovers. This figure shows the contribution of spillovers to the volatility of fixed-income mutual funds’ flows (Panel a) and returns (Panel b) between 1998 and 2014. We measure the contribution to volatility using a calibrated equilibrium model of fund flows (see Section II of the Internet Appendix for details on the model setup and calibration). We choose the model parameters to match their empirical counterparts and show model-implied volatilities for a range of values of the parameter that measures the intensity of the spillover. Each bar in the figure shows the percentage change in volatility for the corresponding intensity of the spillover parameter relative to a benchmark model with no spillover (in pps). We vary the spillover parameters to match its empirical counterpart for four periods: precrisis (1998–2006), crisis (2007Q4–2008H1), crisis (2008H2–2009H1), and postcrisis (2010–2014), as well as for a counterfactual value that is twice as large as that of crisis (2007Q4–2008H1). For each period, we show four sets of volatilities that are calculated based on monthly model-generated series: total volatility, the mean volatility (standard deviation) of monthly fund flows and returns; within-fund volatility, the mean time-series volatility (standard deviation) of monthly fund flows and returns; and quarterly and annual within-fund volatility, the mean of three-month and 12-month time-series volatility (standard deviation) of monthly fund flows and returns, respectively. For example, the value of 22.46 for the total volatility of returns in the post-crisis period in Panel A means that the total volatility of fund returns in the post-crisis period was about 22% higher due to spillovers, which corresponds to an increase from 1.4% in the benchmark to 1.7%. (Color figure can be viewed at wileyonlinelibrary.com)

V. Conclusion

The role of nonbank intermediaries in debt markets has attracted increasing attention in the wake of the 2008 to 2009 financial crisis and the ensuing rapid growth of “shadow banking” institutions such as fixed-income funds. To better understand the sources of run-like fragility that emanate from the asset management sector, we use rich microdata on individual fund flows, returns, and holdings together with a novel approach to measure network linkages across funds. We show that powerful spillover effects arise among funds that hold the same assets, with fire sales induced by investor redemptions hurting peer funds’ performance and flows, leading to further asset sales that have a negative bond price impact.

The approach developed in this paper offers a useful take on fire-sale externalities in debt markets, a question that has not previously been the subject of systematic empirical testing despite the fact that asset fire sales and costly liquidation have been central ideas in financial economics at least since Shleifer and Vishny (1992). There are several avenues along which our approach can be extended. First, while we take a step in the direction of constructing measures of a fund family’s vulnerability to system-wide flow pressures, more can be done to extend the framework for policy evaluation of alternative financial stability tools.

Second, it would be interesting to study fire-sales spillovers in a more explicit structural setting. A structural extension would allow for quantitative evaluation of policy counterfactuals such as the stress-testing scenarios that are now routinely used for banks and have been under consideration by the FSOc for asset managers. In addition, it would allow for welfare evaluation of the effectiveness of monetary policy as a financial stability tool (Stein (2012)) and of other policy measures aimed at reducing fire-sale spillover risk, such as imposing exit fees on open-end funds that are related to the illiquidity of the funds’ assets.

Third, the framework we introduce could be extended to study additional mechanisms that may lead to strategic complementarities in fund flows, including, for example, relative performance evaluation features in fund managers’ compensation contracts (Feroi et al. (2014)) or the concave relation between fixed-income flows and performance (Goldstein, Jiang, and Ng (2017)). Finally, while we provide initial evidence on the financial stability implications of spillovers, clearly more remains to be done. For example, our findings raise the intriguing possibility of cross-sector spillovers from mutual funds to other large players in debt markets, such as insurance companies. Also, extending the analysis to an international setting would help us understand the extent to which fire-sale spillovers contribute to comovement across countries and global volatility in debt markets, questions that have received increasing attention after the “taper tantrum” episode in the summer of 2013.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.

Replication Code.